

Nim1-related kinases regulate septin organization and cytokinesis by modulating Hof1 at the cell division site

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eLife Assessment

This **valuable** study determines the functional requirements for localization and activity of *S. cerevisiae* septin-associated kinases using in vivo imaging, in vitro and in vivo protein-protein interaction assays, and an instructive in vivo "tethering" approach. In addition to confirming previous results, the study offers evidence that the septin-associated kinases may directly interact with the contractile ring machinery. Although the experiments appear to have been conducted correctly, the quantitative analysis of some experiments is **incomplete** and should be improved to strengthen the conclusions.

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Abstract

The septin scaffold recruits and organizes the actomyosin ring (AMR) components, thus ensuring faithful cytokinesis. The septin-associated kinases - Elm1, Gin4, Hsl1, and Kcc4 are believed to stabilize the septins at the bud neck, but the underlying mechanisms are largely unknown. Here, we present a comprehensive, quantitative analysis of these four septin regulatory kinases and reveal major roles for Elm1 and Gin4 in septin stability. We find that Elm1 and Gin4 play an overlooked role in actomyosin ring organization and constriction. We report that Gin4 kinase directly interacts with F-BAR protein Hof1 via its C-terminal membrane-binding domain and may be involved in proper organization and anchoring of AMR component Hof1 at the bud neck, representing an unappreciated mode of regulation of cytokinesis by the septin kinase network. We also show that Gin4 controls septin organisation and AMR constriction in a kinase-independent manner similar to Elm1. We have also performed an extensive GFP-GBP-based tethering screen in $\Delta elm1$ and $\Delta gin4$ cells and found an important role for Hsl1 in maintaining septin organisation and cell shape in coordination with Elm1, Gin4, and Kcc4. Furthermore, our data indicate that Hsl1 acts downstream of Elm1, with its membrane-binding KA1 domain being critical for its function. Together, these findings reveal new insights into modes of cytokinesis regulation by kinases Gin4 and Elm1

and highlight a redundant role for Hsl1 in controlling septin organization and cytokinesis, revealing the in-built adaptability of the septin kinase network in *S. cerevisiae*.

Introduction

Septins are an evolutionarily conserved family of GTPases, although recent studies have revealed that some septin proteins have lost the ability to hydrolyze bound GTP during evolution^{1,2}. These filament-forming proteins assemble into higher-order cytoskeletal structures, such as filaments, rings, bundles, gauzes, etc., on the membrane^{3–7}. Septin structures act as scaffolds for assembling and organizing various proteins required for cytokinesis and cell division across organisms^{8,9}. Septins act as a diffusion barrier at the cell division site in fungi and animals, contributing to contractile actomyosin ring (AMR) assembly, maturation, and constriction during cytokinesis^{10–13}. However, studies have also highlighted that redundant mechanisms cooperatively regulate AMR components, suggesting that the septin diffusion barrier is not an essential requirement for normal cytokinesis^{9,14}. The budding yeast *Saccharomyces cerevisiae* is an excellent model for studying the relationship between septins and cytokinesis. *S. cerevisiae* expresses five mitotic septins: Cdc3, Cdc12, Cdc10, Cdc11 and Shs1^{15–19}, which assemble into octameric complex to form filaments or septin rods that eventually organize into diverse higher-order architectures on the membrane^{3,20–21}.

The presence of Cdc11 and Shs1 at the terminal positions of the octameric unit determines the type of higher-order septin structures in budding yeast, with Cdc11 promoting septin bundling and Shs1 facilitating the formation of septin rings^{22,23}. The septin filaments span the mother bud neck as a stable “hourglass” structure, providing a scaffold for various proteins to assemble into an actomyosin ring^{7,9,10}. The hourglass septin scaffold is remodeled into double rings upon mitotic exit and cytokinesis onset^{24–26}. This allows AMR to anchor properly to the membrane, leading to normal constriction and centripetal septum deposition between the dividing cells^{10,27–30}. The exact molecular events that regulate the hourglass-to-double-ring (HDR) transition are still not known, and the mitotic exit network (MEN) is believed to play a crucial role in initiating this transition^{25,26}.

Septin-associated kinases regulate the dynamics of septins by associating with the septin hourglass structure from the G1 phase and leaving the bud neck at the time of the HDR transition³¹. Septin-associated kinases are one of many key candidates believed to drive the drastic septin rearrangement at the onset of cytokinesis^{32,33}. Four major conserved septin-associated kinases – Elm1^{33–36}, Gin4^{32,37–43}, Hsl1^{44–49}, and Kcc4^{50,51} are thought to have distinct and common regulatory effects on septin stability and organisation^{32,33,52–54}. Despite their involvement in stabilizing septins, a comprehensive understanding of their crosstalk and coordinative effects on septin organisation and cytokinesis in *S. cerevisiae* is just beginning to be understood³².

In this study, through systematic analysis of these kinases on septin stability and actomyosin ring-mediated cytokinesis, we find a role for the Elm1 and Gin4 kinases in maintaining AMR organisation and constriction. The AMR components are disorganized upon loss of Elm1 and Gin4, and AMR displays asymmetric constriction, leading to abnormal cytokinesis defects. We show that the control of Gin4 over AMR organisation is independent of its kinase activity and may rely directly on its interaction with the septin-binding protein Hof1, representing a direct crosstalk between the septin kinase network and cytokinetic machinery. We find that artificial tethering of Hsl1 alone or in combination with Gin4/Kcc4 to the bud neck rescues morphological defects exhibited by $\Delta elm1$ cells. These results suggest an inherent redundancy in controlling septin architecture and cell polarity in budding yeast where Hsl1 kinase can compensate for the role of Elm1. Overall, our work unveils 1) a kinase-independent role of Gin4, 2) a novel physical association between the septin-associated kinase Gin4 and F-BAR domain containing AMR protein

Hof1, involved in direct regulation of AMR organization and cytokinesis, 3) an previously uncharacterized role for Hsl1 kinase in regulating septin scaffold and AMR dynamics in the absence of major kinases Elm1 and Gin4 to ensure proper cell cycle progression and cytokinesis in *S. cerevisiae*.

Results

Dynamics of Septin-associated kinases and their defects during the cell cycle

S. cerevisiae contains four major septin-regulatory kinases known to localize to the mother-bud neck – Elm1, Gin4, Hsl1, and Kcc4, contributing to septin stability and morphogenesis checkpoint during the cell cycle^{33,52}. Three of these kinases, commonly referred to as Nim1-related kinases: Gin4, Hsl1, and Kcc4, consist of membrane-binding Kinase-Associated (KA1) domain at the C-terminus (Fig. 1A). The KA1 domain shows affinity to acidic phospholipids (phosphatidylserine) on the plasma membrane and is a requisite for bud neck targeting^{40,42,44,50}. To understand their temporal kinetics and localized accumulation at the bud neck septin scaffold, we performed time-lapse imaging of GFP-tagged kinases in cells co-expressing the septin Cdc3-mCherry. All four kinases showed gradual recruitment to the septin hourglass scaffold starting from the G1 phase, reaching their peak intensity during M-phase (Fig. S1A-S1D, Movie-S1). Elm1 and Kcc4 were present at lower abundance at the bud neck (Fig. S1A, S1D) compared to the higher abundance of Gin4 and Hsl1, as observed from their fluorescence intensities (Fig. S1B, S1C). At the onset of cytokinesis, the septin hourglass scaffold undergoes remodelling into a double-ring structure, and we observed that all four kinases left the bud neck just before or during the HDR transition (Fig. S1E, S1F, Movie-S1). Interestingly, Elm1 and Kcc4 signal intensity started decreasing ~4-6 mins before the HDR transition (Fig. S1E, S1F). In contrast, Gin4 and Hsl1 signal intensity started decreasing just before or during the HDR transition, suggesting them to be a possible trigger for septin remodelling (Fig. S1E, S1F). These observations suggest a defined temporal order of removal for these kinases from the bud neck, which may depend on the interdependency of their localizations on one another. It is also worth noting that since Gin4 and Hsl1 leave the hourglass septin scaffold very close to the HDR transition (Fig. S1E, S1F), they may act as molecular signals for the transition and a trigger for crosstalk of cytokinesis. Overall, the above data shows that the four septin kinases are majorly associated with the hourglass septin scaffold with similar recruitment but distinct removal kinetics, suggesting that they may play both overlapping and non-overlapping roles in septin hourglass stability and HDR transition.

Septin-associated kinases Elm1 and Gin4 majorly regulate septin stability, HDR transition, and cell morphology

Septin rearrangement is believed to be partly governed by post-translational modifications like phosphorylation, ubiquitination, acetylation and sumoylation^{31,37,55-60}. Previous studies have extensively highlighted the regulation of kinase-mediated modification on septin remodelling^{33,53,61,62}. Septin-associated kinases are believed to play an essential regulatory role in stabilizing the septin hourglass structure and maintaining a timely transition to the double-ring structure upon mitotic exit in *S. cerevisiae*^{31-34,52}. To understand the extent to which different septin kinases affect cell physiology and septin organisation, we characterized cell morphology, cell growth, and septin organisation in wildtype cells and cells harboring deletions for these septin-regulatory kinases – Elm1, Gin4, Hsl1, and Kcc4. We observed that cells exhibited elongated and clumped morphologies in the $\Delta elm1$ and $\Delta gin4$ backgrounds, respectively, at all temperatures tested (Fig. S2A, S2B). $\Delta elm1$ cells displayed a more severe defect, with nearly 100% of cells, whereas $\Delta gin4$ cells showed defects in over 70% of the population (Fig.

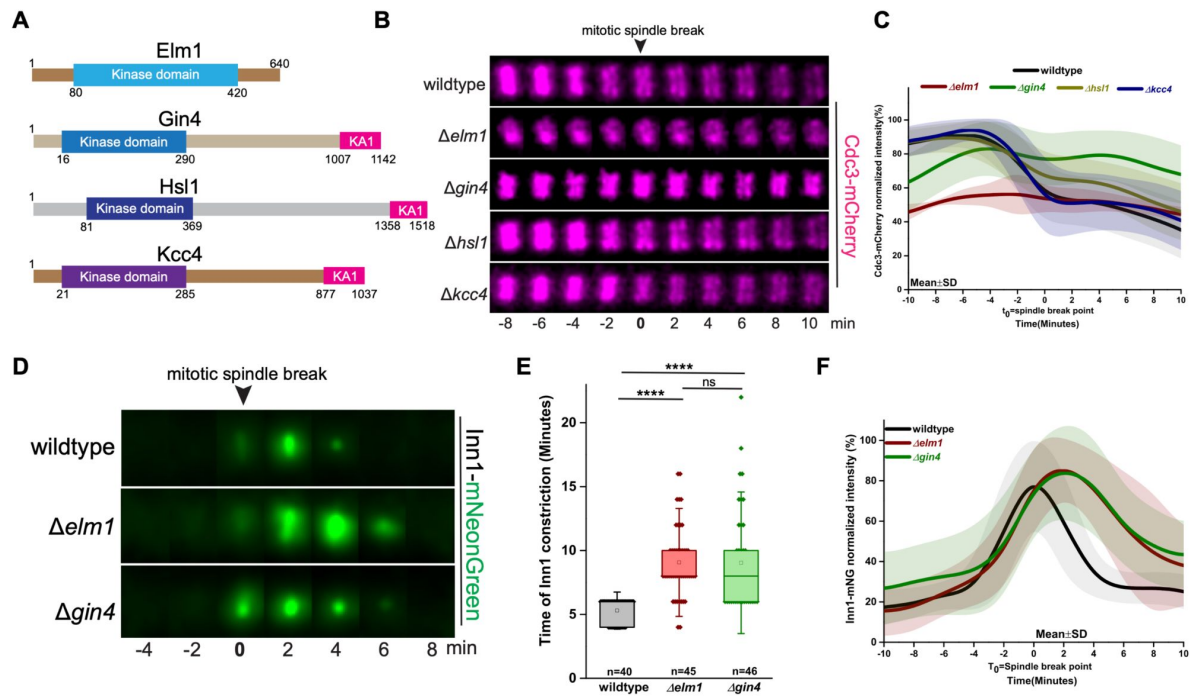


Figure 1.

Septin organization and actomyosin ring dynamics is defective in $\Delta elm1$ and $\Delta gin4$ cells.

(A) Domain-level architecture of Elm1, Gin4, Kcc4 and Hsl1. The KA1 domain represents the membrane-binding domain. (B) Representative time-lapse images of Cdc3-mCherry in kinase deletion strains (t_0 =spindle breakpoint). (C) Plot showing the kinetics of Cdc3-mCherry in indicated strains of (B) ($n > 25$ cells/strain). (D) Representative time-lapse images of Inn1-mNG in wildtype ($n=21$), $\Delta elm1$ ($n=13$) and $\Delta gin4$ cells ($n=23$). (E) Quantification for Inn1-residence in indicated strains of (D). ****: $p < 0.0001$, ns: $p > 0.05$ (F) Plot of normalised signal intensity of Inn1-mNG in indicated strains of (D); a population of cells showing asymmetric constriction in $\Delta elm1$ and $\Delta gin4$ is plotted (t_0 =spindle breakpoint).

S2A, **S2B**). *Δhsl1* cells showed a fraction of cells with mild morphological defects, which increased with temperature, while no alterations in morphology were observed in *Δkcc4* cells (**Fig. S2A**, **S2B**). Spot assay analysis showed a similar trend where *Δelm1* and *Δgin4* showed growth sensitivity at all temperatures tested, while *Δhsl1* exhibited growth sensitivity at higher temperatures and *Δkcc4* cells grew identical to wildtype (**Fig. S2C**).

The core septin Cdc3 fused to mCherry was used to visualize the septin structures during the cell cycle⁷⁷. Cdc3-mCherry localized to the emerging bud site in wildtype, *Δelm1*, and *Δgin4* cells but was also seen to be mislocalized to the bud tip in ~100% *Δelm1* and ~70% *Δgin4* cells (**Fig. S2D**, **S2E**, **Movie-S2**). In wildtype cells, septin formed an hourglass structure at the bud neck, which subsequently split into double rings coinciding with spindle breakage and mitotic exit (**Fig. 1B**). The septin hourglass architecture appeared abnormal in *Δelm1* and *Δgin4* cells (**Fig. 1B**, **1C**) and also exhibited defective HDR transition (**Fig. 1B**, **1C**, **Movie-S2**). *Δelm1* cells showed a stronger defect in septin organisation than *Δgin4* cells (**Fig. 1B**, **1C**, **S2D**, **S2E**), suggesting a more prominent role for Elm1 in maintenance of septin filament architecture, consistent with previous literature³³. Cells lacking the kinases Hsl1 and Kcc4 showed normal septin organisation and HDR transition similar to wildtype cells (**Fig. 1B**, **1C**, **S2D**, **S2E**). However, approximately 15% of *Δhsl1* cells displayed septin mislocalisation to the bud tip (**Fig. S2D**, **S2E**), indicating that Hsl1 and Kcc4 may also play minor or supporting roles in septin dynamics. These results demonstrate the requirement of Elm1 and Gin4 kinases for proper septin arrangement and dynamics at the cell division site.

Elm1 and Gin4 regulate actomyosin ring organisation and constriction

Next, we investigated the roles of the kinases Elm1 and Gin4 to understand their impact on AMR organisation and constriction. We used Inn1, a plasma membrane ingression protein that couples AMR constriction with primary septum (PS) formation, as a marker for AMR constriction^{63,64}. Inn1-mNeonGreen (mNG) arrived at the bud neck just after spindle breakage and constricted symmetrically as part of the AMR with a residence time of ~5 minutes and eventually disappeared from the bud neck in wildtype cells (**Fig. 1D**, **1E**). In *Δelm1* and *Δgin4* cells, Inn1-mNG was recruited normally but displayed an abnormal constriction pattern and increased duration of constriction (**Fig. 1D**, **1F**, **Movie-S3**). On the other hand, in *Δhsl1* and *Δkcc4* cells, Inn1-mNG recruited normally and constricted, similar to wildtype (**Fig. S3A-S3C**, **Movie-S3**). Inn1 and Cyk3 function parallelly to activate a primary component of PS, Chitin Synthase II (Chs2)^{63,65-67}. Thus, we monitored the constriction pattern and dynamics of Chs2-mNG and observed an increased residence time for Chs2 at the bud neck in *Δgin4* cells (**Fig. S3D-S3F**). The C-terminal domain of Inn1 is necessary for its bud neck localisation and interacts with both F-BAR protein Hof1, which senses and induces membrane curvature^{63,68}, and the Cyk3 protein⁶³. The tripartite complex Hof1-Inn1-Cyk3, referred to as the Ingression Progression Complex (IPC), eventually activates Chs2 for functional PS formation^{63,64,69}. Our observations suggest that loss of septin kinases Elm1 and Gin4 may result in an improper organisation and constriction dynamics of the AMR and possibly subsequent IPC complex formation. To test for AMR/IPC organization, we assessed the organization of the F-BAR protein Hof1, which is known to be associated with septin hourglass and transit to AMR during split ring trigger⁶⁸⁻⁷⁴. Hof1 deletion has been shown to exhibit asymmetric AMR constriction with improper chitin deposition during PS synthesis⁶³. We then analyzed the residence time of Hof1-mNG at the bud-neck after the spindle breakpoint during cytokinesis. Hof1-mNG exhibited a residence time of 10-12 minutes, significantly increasing to >15 minutes in both *Δelm1* and *Δgin4* cells (**Fig. 2A**, **2B**, **Movie-S4**). In wildtype cells, Hof1-mNG appeared as a uniform ring at the bud neck, while in *Δelm1* and *Δgin4* cells, Hof1-mNG signal was present as a disconnected, broken ring in more than 60% of the cells (**Fig. 2C**, **2E**). These observations suggest that Elm1 and Gin4 regulate the proper spatial assembly of different AMR components to ensure robust cytokinesis. The effect on AMR dynamics observed in *Δelm1* and *Δgin4* cells may be an indirect effect caused by defects in septin

organization. However, a previous study has shown that the disruption of the septin scaffold does not affect cytokinesis, suggesting that Elm1 and Gin4 could regulate AMR organization directly and constriction independently of their effects on septin organization^{9,14}. Together, these results reveal an unappreciated role for Elm1 and Gin4 in AMR assembly, organization, and constriction, opening avenues for future investigation.

Gin4 directly interacts with the F-BAR domain of AMR protein Hof1 to regulate proper cytokinesis

To understand how Elm1 and Gin4 kinase affect AMR dynamics, we performed a Yeast Two-Hybrid screen to find novel interacting partners at the bud neck. We found that the Gin4 fragments, KID (290-1142aa) and KA1 (1003-1142aa) but not the full-length protein, interacts with the full length Hof1 protein (**Fig. 2F**). Further, we found that the N-terminal Hof1 fragment (1-350aa) which contains its F-BAR domain and not the other fragments, interact with the Gin4-KA1 domain (**Fig. 2G**, **S3I**). In contrast neither full length Elm1 nor its C-terminal fragment (421-640aa) showed any interaction with Hof1 (**Fig. 2F**). To further validate the interaction between Gin4^{KA1} and Hof1^{N (1-350)} domains, we purified 6His-bdSUMO-Gin4-KA1^(1003-1142 aa) and GST-Hof1-N-terminus^(1-350 aa) fragments from *E. coli* and performed an *in vitro*-binding assay. Strikingly, our results revealed that the N-terminal, membrane binding F-BAR containing domain of Hof1 (1-350 aa) interacts strongly with the C-terminal membrane-binding, KA1 domain of Gin4 (1003-1142 aa) (**Fig. 2H**, **2I**). Overall, these results reveal a direct physical link between the Gin4 kinase and AMR component Hof1, suggesting a potentially novel and direct mode of regulation of cytokinesis by the septin kinase network in *S. cerevisiae*. The involvement of the membrane-binding interfaces of these proteins also suggests a possible coordinative mechanism where Gin4 may regulate Hof1 to control its proper assembly and anchoring to the membrane prior to cytokinesis (**Fig. 2J**). This is supported by the disorganized and irregular Hof1 ring structure observed in Δ *gin4* cells (**Fig. 2C**). Moreover, Δ *gin4* cells show asymmetric constriction (**Fig. 2A**) which has been reported earlier to be a characteristic phenotype of Δ *hof1* cells⁶³. Thus, our data combined with these lines of evidence strongly suggest that Gin4 exerts direct control over AMR dynamics via Hof1.

Gin4 exerts its control over actomyosin ring components in a kinase-independent manner

Elm1 kinase activity is essential for its localisation to the bud neck but is dispensable for its role in septin stability³³. Bni5 is a myosin-recruiting protein that stabilizes and crosslinks septin filaments by binding to the terminal septin Cdc11⁷⁵⁻⁷⁸. Elm1 regulates the septin hourglass structure and stability through its binding partner, Bni5³³. Therefore, we asked if Gin4 kinase activity is required for its functions in septin regulation and cytokinesis. To address this, we complemented Δ *gin4* cells with the kinase-dead *gin4-K48A*^{32,37,79} allele (henceforth referred to as Gin4^{KD}), integrated into the *leu2* locus. Interestingly, Gin4^{KD} localized to the bud neck and complemented the abnormal septin organisation (**Fig. 3A**, **3B**) and AMR constriction dynamics (**Fig. 3C**–**3E**) observed in Δ *gin4* cells. Additionally, Gin4^{KD} restored Hof1-misorganisation defects when expressed in Δ *gin4* cells (**Fig. S3G-S3H**). Gin4 kinase stabilizes septins by phosphorylating its substrate, Shs1⁴³. Hof1 has been shown to exhibit synthetic lethality with Gin4, possibly by regulating septins in a parallel pathway⁷¹. We, therefore, complemented Gin4^{FL} and Gin4^{KD} in cells carrying Δ *gin4* Δ *hof1* double deletion (**Fig. S3J**). Interestingly, Gin4^{KD} rescued the synthetic lethality, similar to Gin4^{FL}, suggesting that the kinase activity of Gin4 is not required to restore viability in the absence of Hof1 (**Fig. S3J**). These findings provide strong evidence to support the idea that Gin4 kinase activity is majorly dispensable for its localisation and functions in septin organization and cytokinesis.

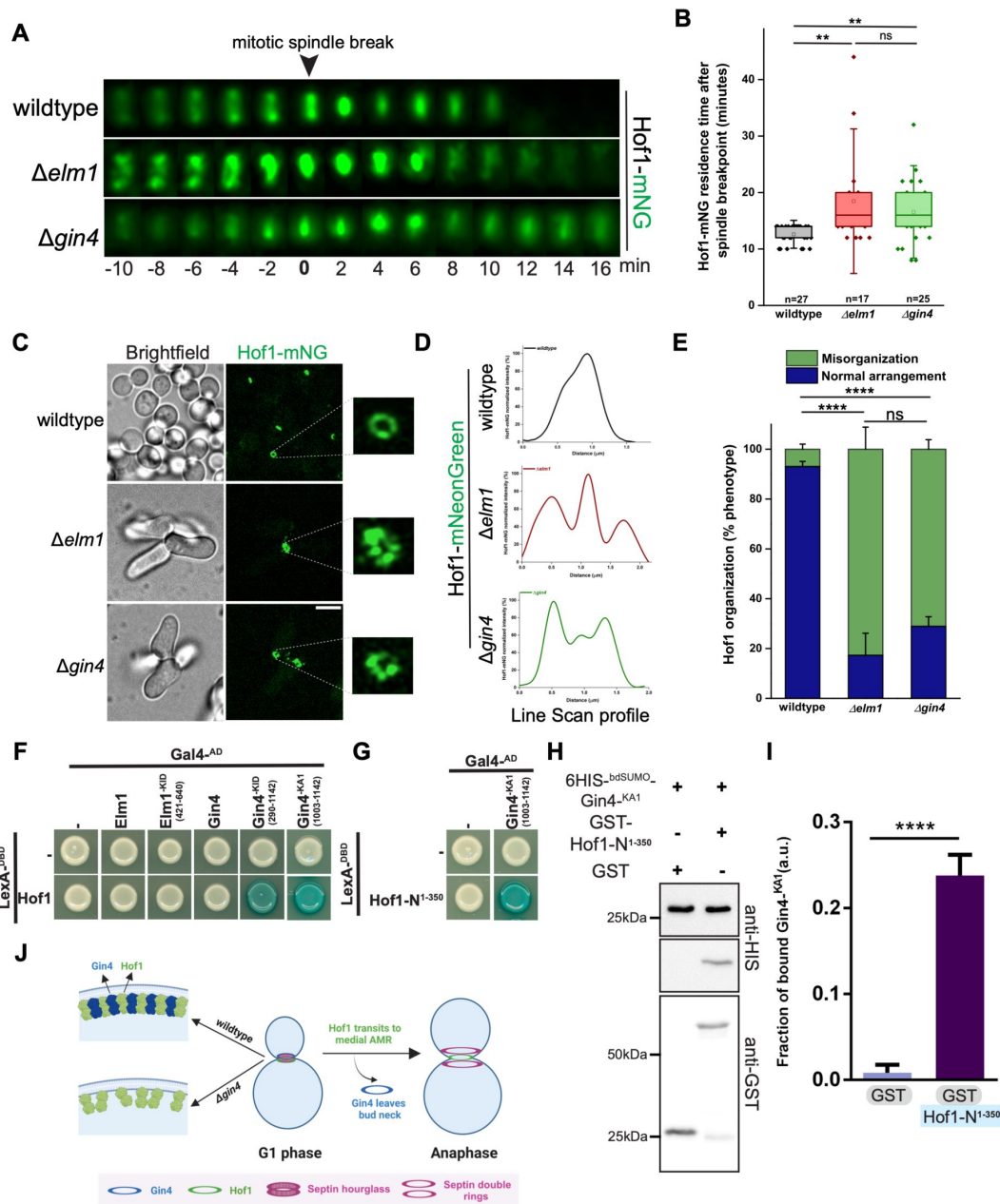


Figure 2.

Gin4 regulates Hof1 organisation and dynamics by interacting with N-terminal membrane binding F-BAR domain.

(A) Representative time-lapse images of Hof1-mNG in wildtype, $\Delta elm1$ and $\Delta gin4$. (B) Quantification for Hof1-residence after spindle breakpoint in indicated strains of (A). **: $p < 0.01$, ns: $p > 0.05$ (C) Representative images showing Hof1-mNG organisation at the bud neck in indicated strains of (A). Scale bar- $5\mu m$. (D) Line-scan profile of (C). (E) Bar graph showing % phenotype for indicated strains of (A) ($N=3$ and $n>200$ /strain). ****: $p < 0.0001$, ns: $p > 0.05$ corresponds to population of cells with disorganised Hof1 at the bud neck. (F) Representative images for Yeast Two-Hybrid assay depicting the interaction of full length Gin4, Gin4-KID and Gin4-KA1 with full length Hof1. (G) Images of Yeast Two-Hybrid showing the interaction between Gin4-KA1 and N-terminal F-BAR containing domain of Hof1. (H) Immunoblot representing the in vitro binding assay for the interaction between 6xHis-bdSUMO-Gin4^{KA1} and GST-Hof1^{N1-350} fragments. (I) Quantification of (H). ****: $p < 0.0001$ (J) Representative model for the possible mode of regulation between Gin4 and Hof1 at the plasma membrane.

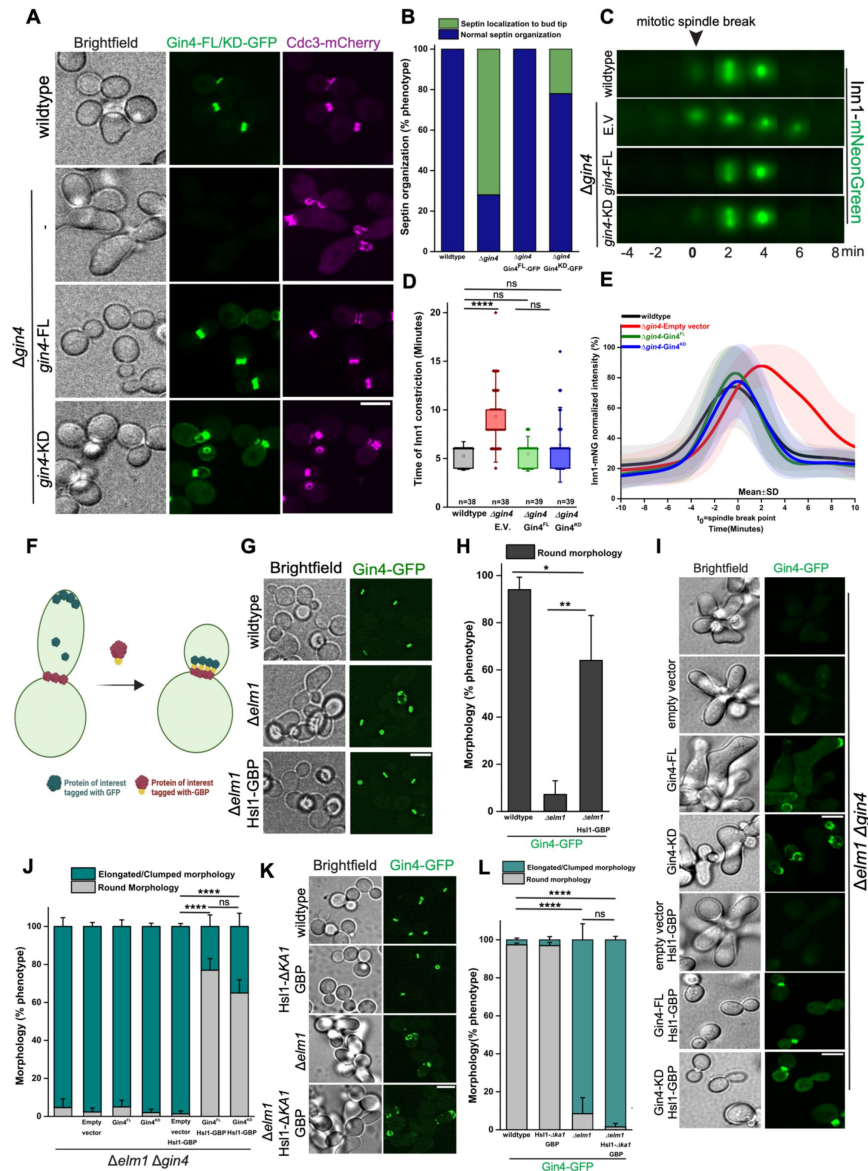


Figure 3.

Artificial tethering of Gin4-GFP to the bud neck, along with its related kinase Hsl1, restores functionality in $\Delta elm1$ cells in a kinase-independent manner.

(A) Representative images showing septin defects in $\Delta gin4$ and rescue by $Gin4^{FL}$ -GFP and $Gin4^{KD}$ -GFP expressed under pTEF promoter -scale bar-5 μ m. (B) Quantitation for septin mislocalisation shown in (A) (The $n > 30$ cells/strain). (C) Representative time-lapse images of Inn1-mNG kinetics in $\Delta gin4$ and rescue by $Gin4^{FL}$ and $Gin4^{KD}$ (t_0 =spindle breakpoint). (D) Quantification for Inn1-residence in indicated strains of (C). ****: $p < 0.0001$, ns: $p > 0.05$ (E) Plot of normalised signal intensity of Inn1-mNG in indicated strains of (C); a population of cells showing asymmetric constriction in $\Delta gin4$ -Empty vector is plotted (t_0 =spindle breakpoint; $n > 20$ cells/strain). (F) A diagrammatic representation of GFP-GBP screen for artificial tethering of Protein of interest to the bud neck, along with its binding partners tagged with GBP. (G) Representative images showing artificial tethering of $Gin4$ -GFP with Hsl1-GBP in $\Delta elm1$. Scale bar-5 μ m. (H) Bar graph showing % phenotype for indicated strains of (G) ($N=3$, $n > 350$ cells/strain). **: $p < 0.01$ (I) Representative images showing $Gin4^{FL}$ -GFP and $Gin4^{KD}$ -GFP tethering with Hsl1-GBP in $\Delta elm1 \Delta gin4$. Scale bar-5 μ m. (J) Bar graph showing % phenotype for indicated strains of (I) ($N=3$, $n > 300$ cells/strain). ****: $p < 0.0001$, ns: $p > 0.05$ corresponds to population of cells with round morphology. (K) Representative images showing artificial tethering of $Gin4$ -GFP with Hsl1- $\Delta ka1$ -GBP in $\Delta elm1$. Scale bar-5 μ m. (L) Bar graph showing % phenotype for indicated strains of (K) ($N=3$, $n > 350$ cells/strain). ****: $p < 0.0001$, ns: $p > 0.05$ corresponds to population of cells with round morphology.

Hsl1 kinase can partially compensate for Elm1 function upon forced tethering to the bud neck

Marquardt et al. (2024) have shown that Gin4 and Elm1 exhibit reciprocal regulation during different cell cycle stages, and their localisation to the bud neck is interdependent^{32,80}. The study also suggests that Gin4 localisation to the bud neck is hampered during late cell cycle stages in *Δelm1* cells³². We independently corroborated these results and observed that Gin4 was mislocalised to the bud cortex during bud emergence and exhibited defective localisation to the bud neck during the onset of cytokinesis in *Δelm1* cells (Fig. S4A-S4D). Next, we asked if restoring Gin4 localisation to the bud neck could rescue elongated morphology in cells lacking Elm1 kinase. To test this hypothesis, we performed a GFP-GBP tethering screen to artificially restore Gin4 localisation at the bud neck in *Δelm1* cells. Interestingly, we observed that tethering Gin4-GFP with the kinase Hsl1-GBP could restore normal morphology in ~60% of *Δelm1* cells (Fig. 3F-3H) while tethering with other binding partners like Shs1^{43,81}, Bni5⁴³, and Anillin-like protein Bud4⁸² did not show any rescue in the phenotype of *Δelm1* cells despite restoration of bud neck localization (Fig. S4E, S4F). To test whether Gin4 kinase activity is required for this observed rescue, we tethered Gin4-GFP and Gin4^{KD}-GFP via Hsl1-GBP in *Δgin4 Δelm1* cells and observed that both constructs restored normal cell morphology (Fig. 3I, 3J). These results suggest that Gin4 tethering to the bud neck via Hsl1 may compensate for Elm1 function in a kinase-independent manner. Finally, we tested whether the membrane-binding Kinase-Associated (KA1) domain of Hsl1 is required for this function and found that Gin4-GFP tethering via Hsl1-*Δka1*-GBP did not rescue cellular morphology of *Δelm1* cells (Fig. 3K, 3L). Together, these observations suggest that (i) restoring Gin4 localisation via Hsl1 is necessary to rescue the morphological defects exhibited by *Δelm1* cells partially, (ii) Gin4 might be forming a complex with Hsl1, which becomes essential in the absence of Elm1 indicating that Elm1 might play a significant role in regulating Gin4-Hsl1 complex. (iii) proper membrane tethering of Hsl1 via its KA1 domain is essential for its function in *Δelm1*.

Intrigued by the rescue specifically observed after tethering Gin4-GFP with Hsl1-GBP, we extended our screen to its related kinases and tethered Hsl1-GFP and Kcc4-GFP via various bud-neck proteins in *Δelm1* cells (Fig. 4A-4D, S5F, S5G). Our results revealed that Hsl1-GFP could restore normal cell morphology in ~30% of *Δelm1* cells when tethered with Shs1-GBP while tethering with septin kinases Gin4-GBP or Kcc4-GBP resulted in a significantly increased percentage of phenotype rescue (~50%), which suggests possible coordination between these kinases (Fig. 4A, 4B). Strikingly, like Gin4-GFP, Kcc4-GFP tethering restored cell morphology similar to wildtype, only when tethered with Hsl1-GBP (Fig. 4C, 4D). However, Kcc4-GFP tethering via different bud neck proteins did not show any rescue of phenotype, but a minor rescue was observed upon tethering with Gin4-GBP, suggesting a coordination between Kcc4 and Gin4 (Fig. S5F, S5G). To support these results, we analyzed the localisation of Hsl1 and Kcc4 in *Δelm1* and *Δgin4* cells. As reported in a previous study³³, Kcc4 exhibited mislocalisation to the bud cortex during the initial phases and exhibited defective localization at the onset of cytokinesis (Fig. S5B-S5E). Recruitment and localisation of Hsl1 to the bud neck are also found to be severely hampered in *Δelm1* cells (Fig. 4E-4G, S5A), suggesting a role for Elm1 in maintaining Hsl1 localization at the bud neck. In contrast, recruitment and localisation of Hsl1 and Kcc4 were similar to wildtype in *Δgin4* cells (Fig. 4E-4G, S5A-S5E). These data suggest that loss of Hsl1 at the bud neck may contribute majorly to the phenotype observed in *Δelm1* cells and consistent with this idea, restoration of Hsl1 localization rescues the observed defects in our experiments. Our results reveal that Hsl1 regulates normal cell morphology, either through downstream signaling events caused by the presence of Elm1 in wildtype cells or by a parallel pathway in coordination with either Gin4 or Kcc4. Thus, Elm1 kinase may act upstream to control the localisation and activation of other regulatory kinases like Hsl1, Gin4 and Kcc4 at the septin

scaffold in a precise, spatiotemporal manner in a network with multiple feedback regulatory links. Overall, the ability to restore normal function by simply rewiring the septin kinase network demonstrates the inherent redundancy and robustness of this network in *S. cerevisiae*.

Hsl1 and its membrane-binding ability are required for the rescue of *Δgin4* cells by artificially tethered Elm1

Elm1 does not localise to the bud neck in the absence of Gin4 and can restore the morphological defects exhibited by *Δgin4* cells upon artificial tethering to the bud neck septin scaffold³². We corroborated these findings independently and found that Elm1 is not recruited to the bud neck in *Δgin4* cells (Fig. S6A–S6D). We find that Elm1-GFP tethering via Shs1-GFP, Bud4-GFP, and Hsl1-GFP could restore normal cell morphology in *Δgin4* cells (Fig. S6E, S6F). Therefore, we asked whether presence of Hsl1 could be necessary for the observed rescue upon Elm1 tethering in *Δgin4* cells. To test this, we tethered Elm1-GFP to the bud neck via Shs1-GFP in *Δgin4* and *Δgin4 Δhsl1* cells. While cell morphology was restored to normal in *Δgin4* cells, no rescue was observed in *Δgin4 Δhsl1* cells, suggesting that Hsl1 is required downstream of Elm1-tethering in *Δgin4* cells (Fig. 5A, 5B). A similar loss of rescue was also observed when Elm1-GFP was tethered to the bud neck in *Δgin4* cells via Hsl1-*Δka1*-GFP, suggesting that Hsl1 acts downstream of Elm1 and its membrane-binding KA1 domain is required for its function (Fig. S6G, S6H). Next, we asked if Elm1-GFP tethering to the bud neck in *Δgin4* cells could rescue other observed cellular defects. To test this, we first assessed the localisation of the type-II myosin Myo1, a core component of AMR, which localizes to the bud neck in a biphasic manner^{10,77}.

We find that Myo1-3xmCherry exhibited mislocalisation to the bud tip in *Δgin4* cells (Fig. S6I). While Myo1-3xmCherry mislocalisation was restored upon Elm1-GFP tethering to the bud neck in *Δgin4* cells, it exhibited misorganization upon deletion of Hsl1 (Fig. S6I). We also assessed Inn1-3xmCherry lifetime in these cells to measure AMR constriction time^{63,64}. We find that Inn1-3xmCherry lifetime is similar to wildtype cells when Elm1-GFP is tethered via Shs1-GFP in *Δgin4* cells but significantly increased when Elm1-GFP is tethered via Shs1-GFP in *Δgin4 Δhsl1* cells (Fig. 5C, 5D). These results suggest that (i) Elm1 tethering to the bud neck restores normal Myo1 localisation and AMR constriction dynamics in the absence of Gin4, (ii) Hsl1 is required for Elm1 to exert its influence over AMR organisation and dynamics after bud neck tethering in *Δgin4* cells, and (iii) Membrane-binding KA1 domain of Hsl1 is required for this activity. Taken together, our results suggest that Hsl1 may play redundant roles along with Elm1 and Gin4 in maintaining septin stability and cytokinesis (Fig. 6). From our observations, one significant role of Elm1 might be in properly recruiting Hsl1 to the bud neck, revealing that Hsl1 may act downstream of Elm1 in this pathway with a partial overlap in their function. The fact that Hsl1 is required for the rescue observed upon Elm1 tethering in *Δgin4* cells also supports the idea that Hsl1 acts downstream of Elm1 in regulating septin stability and cytokinesis.

Discussion

Septins are considered as the fourth cytoskeletal elements due to their extensive structural and functional diversity⁸³. They play critical role in various fundamental processes such as cytokinesis, ciliogenesis, spermatogenesis and phagocytosis etc^{17,84–86}. While their involvement in cytokinesis and cell division has been well-characterized, recent advancements have provided deeper insights into their regulation^{3–5,7,20}. Septin architecture and remodeling have emerged as dynamic areas of investigation, as these processes are spatiotemporally controlled by septin-interacting proteins and post-translational modifications (PTMs)^{31,39,55–57,59,60}. Despite significant progress in the understanding of septin organisation, the mechanisms regulating septins and their modulation by associated kinases in yeast and higher eukaryotes remain incomplete. In *Saccharomyces cerevisiae*, four septin-

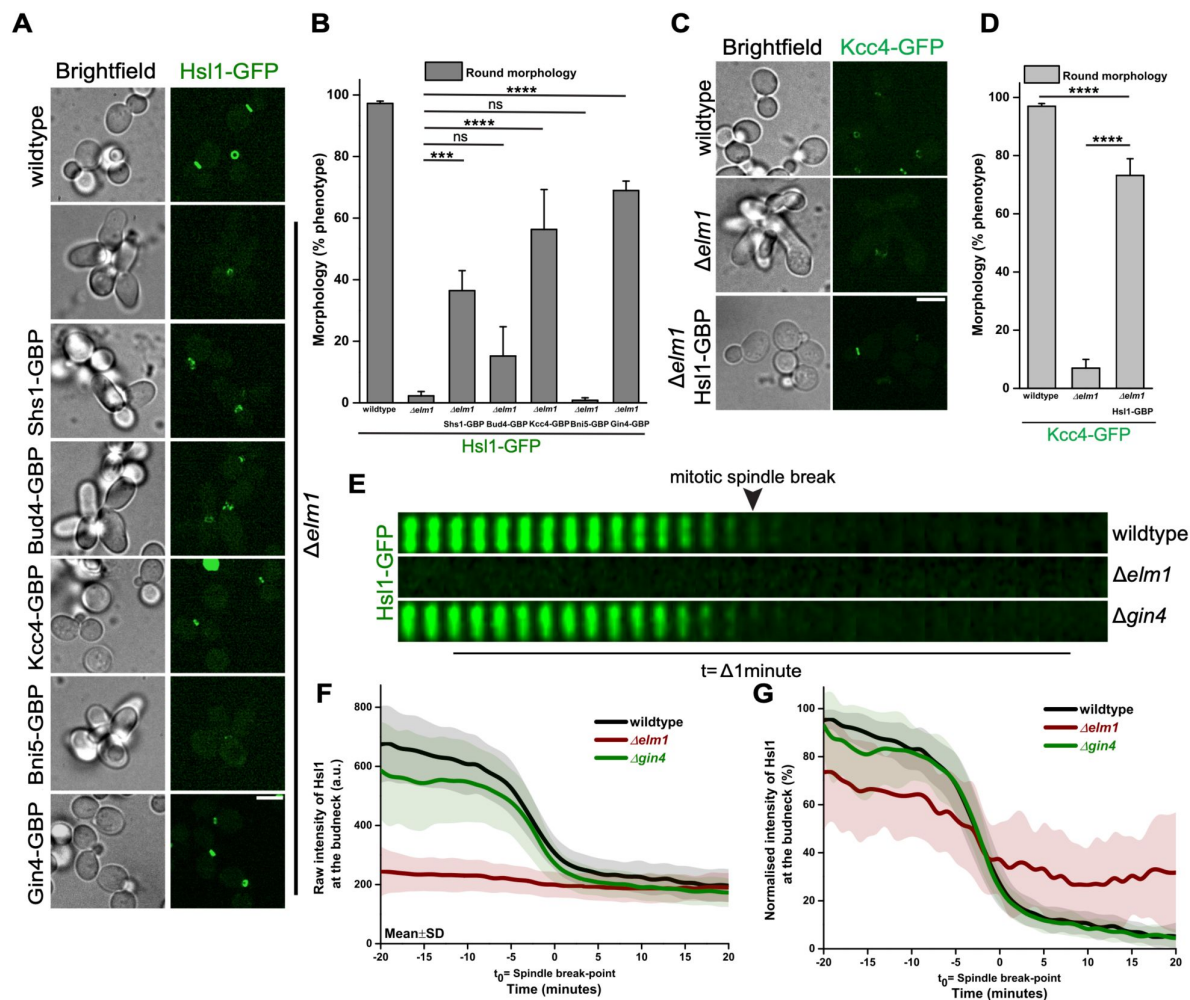


Figure 4.

Restoring Hsl1 localisation to the bud neck can bypass the requirement of Elm1.

(A) Representative images showing artificial tethering of Hsl1-GFP via Shs1-GBP, Bud4-GBP, Kcc4-GBP, Bni5-GBP and Gin4-GBP in $\Delta elm1$. Scale bar=5 μ m. (B) Bar graph showing % phenotype for indicated strains of (A) (N=3 and n>350/strain). ***, p<0.001, ****: p<0.0001, ns: p>0.05 (C) Representative images showing artificial tethering of Kcc4-GFP with Hsl1-GBP in $\Delta elm1$. Scale bar=5 μ m. (D) Bar graph showing % phenotype for indicated strains of (C) (N=3, n>350 cells/strain). ****: p<0.0001. (E) Representative montages of the bud neck depicting the localisation of Hsl1 in $\Delta elm1$ and $\Delta gin4$ cells. (F) Plot of raw intensity of Hsl1-GFP in indicated strains of (E) (t_0 =spindle breakpoint). (G) Plot of normalised signal intensity of Hsl1-GFP in indicated strains of (E) (t_0 =spindle breakpoint).

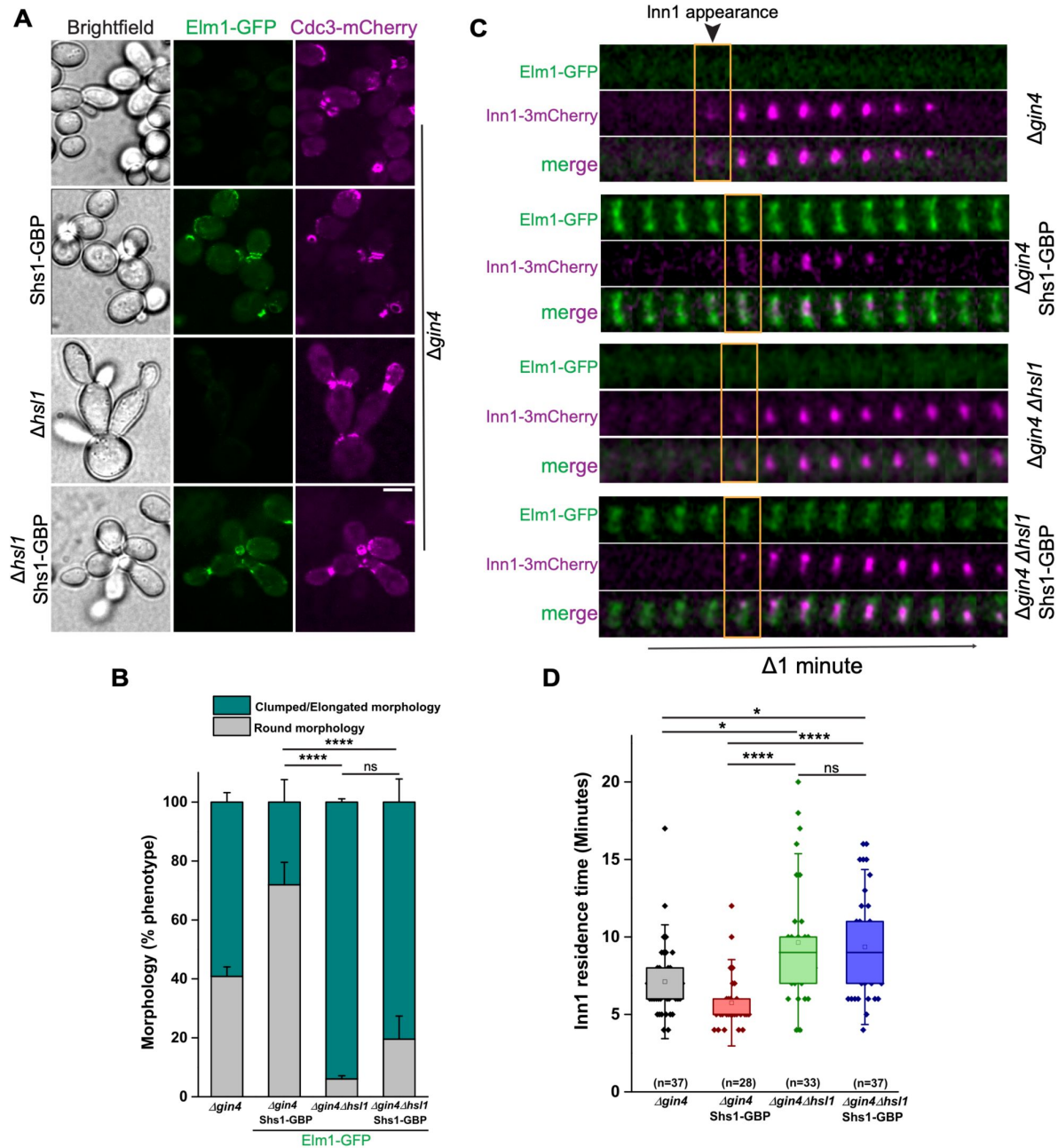


Figure 5.

A non-canonical role of Hsl1-Kinase in regulating septin organisation and AMR dynamics.

(A) Representative images showing septin defects in $\Delta gin4 \Delta hsl1$ with Elm1-GFP tethered to bud neck via Shs1-GBP. Scale bar-5 μ m. **(B)** Bar graph showing % phenotype for indicated strains of (A) (N=3, n>450 cells/strain). ****: p<0.0001, ns: p>0.05 corresponds to population of cells with round morphology. **(C)** Representative time-lapse montages for Inn1-3xmCherry in $\Delta gin4 \Delta hsl1$ with Elm1-GFP tethered to bud neck via Shs1-GBP. **(D)** Quantification for Inn1-residence in indicated strains of (C). *: p<0.05, ****: p<0.0001, ns: p>0.05

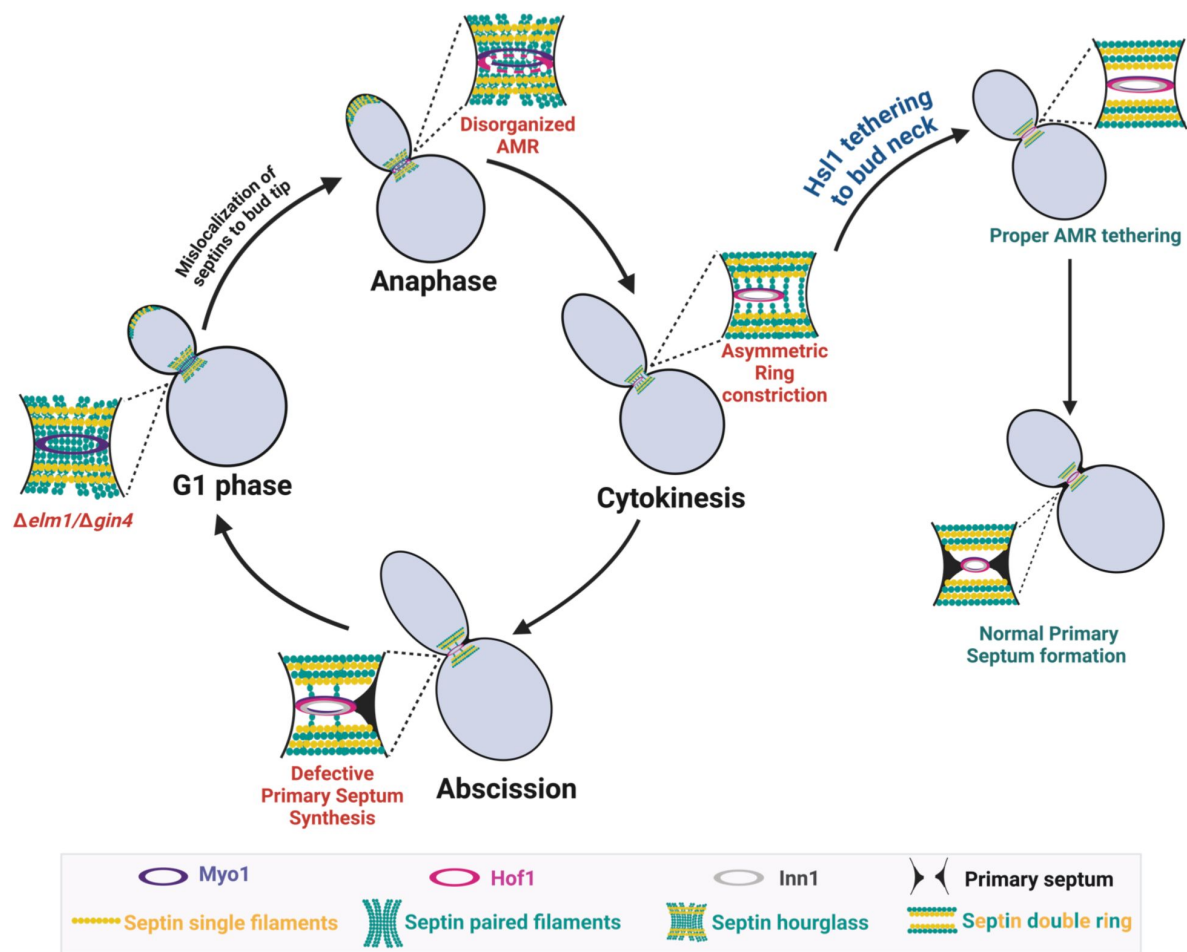


Figure 6.

Representative model for the role of Hsl1 kinase in septin organization and actomyosin ring constriction downstream of Gin4 and Elm1.

associated kinases—Elm1, Gin4, Hsl1, and Kcc4 have been reported to majorly regulate septin-dependent processes^{33,52,87}. Previous studies suggest that Elm1 acts as the most upstream kinase, regulating the Nim1-related kinases (Gin4, Hsl1, and Kcc4) via phosphorylation^{80,87,88}. Interestingly, Elm1 also exhibits kinase-independent roles in septin regulation, potentially through direct interaction with Bni5³³. Gin4 primarily phosphorylates Shs1⁴³, while Hsl1 regulates Swe1 and morphogenesis checkpoint⁴⁸. However, the precise mechanisms through which Nim1-related kinases influence septin architecture—whether directly or via interactions with binding partners—remain poorly understood and requires further investigation.

Septin remodeling marks the onset of actomyosin ring (AMR) constriction and cytokinesis^{9,10,26}. However, the molecular players regulating the crosstalk between these processes remain largely unknown. Septins are severely mislocalized to the bud cortex in *Δelm1* and *Δgin4* cells, suggesting their role in organizing septins at the bud neck consistent with previous reports^{32,33,52}. The defective dynamics of AMR component proteins, Inn1 and Hof1, in the absence of Elm1 and Gin4 kinases further indicate that these kinases regulate intermediate signaling events during cytokinesis and highlight an underappreciated role in maintaining AMR organization and dynamics (**Fig. 1D–1E**, **2A–2E**).

A key question arising from these observations is whether the crosstalk between the septin kinase network and AMR dynamics is direct and septin-independent or indirect and septin-dependent. Previous work has shown that AMR constriction is unperturbed in septin mutants which affect septin organization⁹, suggesting that the deregulation of cytokinesis observed in our study may be a direct effect of the septin kinases, Elm1 and Gin4, on AMR organization and constriction. To support this idea, we also find that Gin4 kinase interacts with the F-BAR protein Hof1 through Yeast-Two Hybrid and direct in vitro binding assays (**Fig. 2F–2I**). The fact that the membrane-binding domains of Gin4 and Hof1 are involved in this interaction (**Fig. 2G–2H**) suggests that Gin4 activation at the membrane as previously suggested^{42,50} may directly involve or affect Hof1 organization and maturation of the AMR. This also suggests a potential mechanism of tethering of Hof1 to the membrane prior to cytokinesis (**Fig. 2J**). Interestingly, previous work has showed that Gin4 and Hof1 are synthetic lethal and that Hof1 exhibits mislocalization in *Δgin4* cells⁷¹. These results also point towards a coordinative mechanism where septin-associated kinase Gin4 directly crosstalk with an AMR component Hof1 to regulate AMR organization and constriction. To our knowledge, this is the first evidence that physically links the septin kinase network to AMR proteins and highlights a novel regulatory input on AMR assembly in *S. cerevisiae*. Future work is required to delineate the molecular mechanisms underlying this regulation.

Our study shows that the kinase activity of Gin4 is not required for its role in septin organization, AMR constriction, and even for maintaining viability in the absence of Hof1 (**Fig. 3A–3E**, **S3J**). This highlights an important aspect of Gin4 function in cells and is similar to kinase-independent function reported for Elm1³³. This poses the question: What is the role of the kinase activity of Elm1 and Gin4? Our study and a previous study^{32,52} have now established that there are in-built redundancies within the septin kinases (Elm1, Gin4, Hsl1, Kcc4) in terms of function but their recruitment to the bud neck is also interdependent on each other. The kinase activities may play an important role in regulating the signaling that coordinates the precise arrival and spatiotemporal activation of these kinases at the bud neck. These results also raise important questions about post-translational regulatory mechanisms of septin filament organization as a number of studies have shown that these kinases directly phosphorylate septins⁴³. Thus, the role of post-translational regulation on septins will be better understood with future work in this direction.

Although these kinases exhibit distinct functions, they also display interdependence on each other^{32,80}. Recent studies propose a new perspective, revealing reciprocal regulation between Gin4 and Elm1 through direct binding and phosphorylation at different stages of the cell cycle³². An unresolved question over the years has been the difference in phenotype observed in *Δelm1* vs

Δgin4. Δelm1 cells shows a greater percentage of abnormality in cell shape and septin organization as compared to *Δgin4* cells even though Elm1 is also known to be absent from the bud neck in *Δgin4* cells^{32,52}. Our work offers a possible explanation for this by analysis of the temporal kinetics of Hsl1 and Kcc4 in *Δelm1* and *Δgin4* cells, which are perturbed only in *Δelm1* and not in *Δgin4* cells (Fig. 4E–4G, S5A–S5E), possibly explaining why cells lacking Elm1 exhibit more pronounced defects. These findings suggest that the localization of Hsl1 and Kcc4 to the bud neck is crucial for normal function and may contribute to the less prominent phenotype observed due to loss of Gin4, emphasizing and strengthening the redundant roles of Hsl1 and Kcc4 in septin organization.

We also performed an extensive GFP-GBP screens in *Δelm1* cells and found that restoring the localization of Hsl1 with Gin4 and Kcc4 can bypass the requirement for Elm1 during the cell cycle (Fig. 3G, 3H, 4C and 4D) highlighting a functional redundancy among the Nim1-related kinases in achieving successful cytokinesis. Our results also suggest that Hsl1 plays a non-canonical role in regulating septin organization and actomyosin ring dynamics downstream of Elm1 and Gin4, providing novel insights into Hsl1's role in cytokinesis (Fig. 5A–5D, S6I). Therefore, we speculate synergy exists between Elm1 and Gin4 to regulate Hsl1 in a downstream signaling pathway that can regulate septin organization and cytokinesis. The requirement of Hsl1's kinase activity in the absence of Elm1 or Gin4 could offer further understanding of the molecular mechanisms underlying Hsl1's regulation in future studies. Strikingly, the inability to rescue this defect upon deletion of the membrane-binding KA1 domain of Hsl1 (Fig. 3K, 3L, S6G, S6H) highlights the importance of membrane-binding for the interplay and activation of these kinases. How membrane binding facilitates Hsl1 function remains to be explored. The underlying molecular mechanisms and actual cause-effect relationships of how Hsl1 and other kinases regulate septin organisation at the filament-level remain to be understood and approaches such as Platinum-replica electron microscopy (PREM) and invitro reconstitution assays will be key to generate such insights in future studies^{21,33,68}. It promises to reveal critical insights into the modes of septin regulation through the coordinated action and signaling inputs from multi-septin kinase modules across eukaryotes.

In summary, our study provides a comprehensive quantitative analysis of the cellular functions of septin-regulatory kinases in septin organization and cytokinesis. Notably, we identified Elm1 and Gin4 as key players in septin regulation, uncovering their previously uncharacterized roles in actomyosin ring organization and constriction. These findings pave the way for future investigations into the intricate molecular mechanisms and cross-regulatory interactions within the septin-kinase network and with actomyosin ring proteins during the cell cycle in *Saccharomyces cerevisiae* and other related organisms where septins and related proteins play a role in cell division.

Materials and methods

Strain construction

All the yeast strains in this study were constructed using the *S. cerevisiae* wildtype strains ESM356 and YPH499 derived from S288C genetic background. The strain number and the genotype are listed in supplementary table S1. The C-terminal epitope tagging, and endogenous gene deletions were carried out based on the PCR-based integration strategy^{89,90}. All the strains were cultured at 23°C with 180rpm. Synthetic complete (S.C.) media lacking tryptophan, uracil, leucine and histidine were used to select auxotrophic recombinants. Antibiotic selections for endogenous gene manipulation were done on YPD agar plates supplemented with 200μg/ml G418 (Sisco Research Laboratories Pvt. Ltd.; catalogue no. 58327), 100μg/ml of hygromycin-B (Sisco Research Laboratories Pvt. Ltd.; catalogue no. 67317) and 100μg/ml of nourseothricin (Jena Biosciences; catalogue no. AB-102). To visualize the septin dynamics, Cdc3-mCherry, YIP211-cdc3-mCherry⁷⁷

and YIP204-cdc3-mCherry⁹¹ plasmids were cut with BglII (catalogue no. R0144S; New England Biolabs) and was integrated into CDC3 locus. Cell cycle dynamics were visualized using a yeast expression vector with mRuby2-Tub1 under His3 promoter (pHIS3p:mRuby2-Tub1+3'UTR::Ura3)⁹² and pAFS125-GFP-TUB1⁹³, which were cut with ApaI (Catalogue no. R0114S; New England Biolabs) and StuI (catalogue no. R0187S; New England Biolabs) respectively and were integrated into *ura3* loci. Yeast transformation for epitope tagging and endogenous gene deletions were performed using a lithium acetate (LiOAc)-based protocol described⁹⁴.

Oligonucleotides and Plasmids

All the oligonucleotides used for endogenous gene tagging and deletion in this study are listed in the supplementary table S2. The plasmids used and generated for this study are listed in the supplementary table S3. For a generation of clones used in **Fig. 3A**, **3B** the Gin4^{FL} was amplified from the endogenous locus and was integrated into pRS305 under the pTEF promoter. The ligation reaction was performed using NEB Hifi-Builder (New England Biolabs; cat.no: E2621L). Catalytically dead Gin4^{KD}(K48A) was constructed based on the site-directed mutagenesis strategy using the oligos (P855, P856, P857 & P1024). The kinase-dead mutant was confirmed correct by sequencing. To construct Gin4^{FL} and Gin4^{KD} used in **Fig. 3C**, **3E**, **S3G**, **S3H** and **S3J** the Gin4 gene fragment was amplified from genomic loci with the endogenous promoter using PCR and was ligated into a pRS305 integration vector cut with BamHI-HF (Catalogue no. R3136S; New England Biolabs).

Live-cell imaging

The cells were inoculated in S.C. media and incubated overnight at a 23°C. The secondary culture was diluted to log-phase (Optical density 0.4 to 0.6) from the overnight saturated culture. The 6% Concanavalin A type 6 (Sigma-Aldrich; catalogue no. C2010) was coated onto a confocal dish (ibidi GmbH-catalogue no. 81218-200; Cellvis-catalogue no. D35C4-20-1.5-N) and was incubated at 23°C incubator. The secondary culture cells were pelleted at 3200 rpm for 3 minutes. The cells were washed in S.C. media to remove non-adherent cells and subjected to live-cell imaging.

The laser point scanning confocal microscope Olympus FV-3000, equipped with high-sensitivity GaAsP photomultiplier tube (PMT) detectors and solid-state lasers (488 nm and 561 nm), was used to acquire time-lapse imaging for **Fig. S1A**, **S1E**. The images were acquired at a one-minute time interval with 0.5µm step size and seven z-slices for 2-3 hours. The raw images were three-dimensional (3D) deconvolved in Olympus CellSens Dimension (3.1) software. Maximum-intensity projection images were used for representation.

The Olympus IX83 widefield microscope equipped with a CoolLED PE-4000 LED light source and Prime-BSI sCMOS camera was used for acquiring images represented in **Fig. 1B**, **1D**, **2A**, **3C**, **S2D**, **S3A** and **S3D**. All the images were acquired in a 100x oil-immersion objective with 1.45 N.A., 1µm step-size and three z-slices at a 2-minute time interval for 60-150 minutes. The raw images were processed for 3D deconvolution using Olympus CellSens Dimension (3.1) software. Maximum-intensity projection images were used for representation.

The images for Hof1 mis-organization and artificial tethering experiments in **Fig. 2C**, **3G**, **3K**, **4A**, **4C**, **S3G**, **S4E**, **S5F**, **S6E** and **S6G** were acquired using the Oxford Andor Dragonfly 502 spinning disc confocal system equipped with a fully motorized Dmi8 inverted setup. The raw images were captured in 100x oil immersion objective with 0.5µm step-size, 11 z-slices in Andor Sona sCMOS camera and deconvolved using Andor Fusion software. The 3D-deconvolved and max-projected images were used for representation.

Olympus IX-83 Laser scanning spinning disc Yokogawa-W1 microscope was used to capture **Fig. 3A**, **3I**, **5A** and **S6I** images. The raw images were acquired in a 100x oil-immersion objective with 1.45 N.A., using 1µm step-size and five z-slices. The time-lapse imaging for **Fig.**

4E [4E](#), 5C [5C](#), S4A [S4A](#), S4B [S4B](#), S5A [S5A](#)-S4C [S4C](#), S6A [S6A](#), S6B [S6B](#) was done at a one-minute time interval in 60x objective with 1.42 N.A. The images and time-lapse montages were 3D-deconvolved in Olympus CellSens Dimension (3.1) software. 3D-Deconvolved and maximum intensity projection images were used for representation.

Protein accumulation kinetics quantification and analysis

The live-cell imaging data acquired was opened in Fiji (ImageJ2) image analysis software⁹⁵ using the protocol established⁹⁶. In brief, a region of interest (ROI) was drawn around the bud neck of the sum-intensity projected image. The extracted fluorescent intensity values were corrected using background values. All the individual fluorescent intensity values obtained in an Excel sheet at each time point were normalized using the minimum and maximum values according to the formula below:

$$\text{Normalized value} = \frac{(\text{Raw value} - \min(\text{distribution}))}{(\max(\text{distribution}) - \min(\text{distribution}))}$$

The data points ranging between the -20 and +20 time points were separately extracted for the 'n' number of cells indicated with each graph, respectively. Mean, and standard deviation was calculated. A line plot was generated using Origin-Pro (2015, Sr2, 69.2.272; Origin Lab, USA) image analysis software to represent the temporal accumulation kinetics of the protein of interest. The small bud is considered as a T_0 for quantifications involving bud-emergence. The complete drop in Cdc3-mCherry intensity is considered as HDR transition in **Figure S1E**. The Spindle breakpoint is considered as T_0 or reference time-point for measuring Hof1, Inn1 or Chs2 temporal dynamics during cytokinesis.

Quantitative analysis of septin and Hof1 misorganization was done manually by counting the number of cells exhibiting septin localisation at the bud cortex during bud emergence and the defective/discontinuous Hof1 ring at the bud neck.

Time-lapse imaging of AMR components was done to analyze the residence time. The time was calculated as the difference between the first-time point where the signal initially appears and the final time point where the signal diminishes completely from the bud neck.

Yeast Two Hybrid Assay

Genes or gene fragments of interest were amplified and cloned into NotI restriction enzyme (NEB; Catalogue No. R3189S) cut pMM5S and pMM6S plasmids consisting of DNA Binding Domain of LexA (Lex^{DBD}) and Activation Domain of Gal4 (Gal4^{AD}) respectively. The clones were transformed into the yeast strains SGY37 (MATa) and YPH500 (MATalpha). The transformants were selected on S.C. media lacking Histidine and Leucine, respectively. Mating was performed on YPD plates and was incubated at 30°C for 2 days followed by replica plating on SC-His-Leu double selection plates. The plates were incubated for 2 days at 30°C, subsequently overlaid with freshly prepared X-Gal mix (500 mM sodium phosphate buffer (pH7.0), 10% SDS, 1 M KCl, 1 M MgCl₂, 0.04% X-gal and 0.4% low melting agarose) for detection of β -galactosidase activity^{97,98}. Interaction of LexA^{DBD}-Protein of interest A with Gal4^{AD}-Protein of interest B resulted in the expression of the *lacZ* gene encoding the β -galactosidase, which converted X-gal to a product with blue colour⁹⁹. The plates were scanned and documented after 14-16 hours of incubation with overlay mix.

Protein expression, purification and In vitro-Binding Assay

The N-terminal fragment of Hof1 containing the F-BAR domain (1-350 aa) and KA1 fragment of Gin4 (1003-1142 aa) were PCR amplified from the *S. cerevisiae* genomic DNA and was cloned into pGEX-5X1 cut with EcoRI restriction enzyme (Catalogue No. R3101S) and pET28a-6His-bdSUMO vector cut with NdeI restriction enzyme (Catalogue No. R0111S) respectively using NEB Hifi Builder. Both the plasmids were transformed into BL21-DE3 *E.coli* competent cells and was

selected on Luria Bertani (LB) plate containing Kanamycin (50mg/ml) for pGEX-5X1-Hof1^{N(1-350)} and Ampicillin (100mg/ml) for pET28a-6His-bdSUMO-Gin4^{KA1}. The LB broth with Kanamycin or Ampicillin was used to grow primary culture overnight at 37°C with constant shaking at 200 rpm. The primary culture was diluted to 0.1 O.D. in 500ml LB broth with Kanamycin or Ampicillin. The protein expression was induced with 0.5 mM isopropyl β-D-1-thiogalactopyranoside (IPTG) once the O.D. reached log phase (0.4-0.5). The culture was incubated at 30°C with shaking of 200 rpm. for 3-4 hrs. The cells were pelleted down 4°C, 7000 rpm for 20 min. Media was aspirated and the pellets were washed with cold phosphate buffered saline (PBS) containing 1mM phenylmethylsulfonyl fluoride (PMSF) and was stored at -80°C.

Purification of 6His-bdSUMO-Gin4^{KA1}

Pellets were kept on ice for 10 min. and was subjected to lysis with the Lysis Buffer (50 mM Tris pH-8, 200 mM NaCl, 10% glycerol). The lysed pellets were clarified at 4°C and 15,000xg. The clarified lysate was mixed with Ni-NTA agarose resin (cat. no. 786-940, G-Biosciences), washed with wash buffer (50 mM Tris pH-8, 200 mM NaCl, 10 mM Imidazole) and was incubated at 4°C for 2 hrs. The lysate with Ni-NTA resins were then transferred to poly-prep chromatography columns (cat. no. 731-1550, BIO-RAD laboratories Inc.), and was subjected to gradient elution with increasing concentration of imidazole. The eluted protein fractions were buffer exchanged with storage buffer (50 mM Tris-HCl pH 7.4, 2000 mM NaCl, 1 mM DTT and 10% glycerol) using PD Midi Trap G-10 Sephadex columns (GE Healthcare, cat. No. GE28-9180-11). The purified proteins were flash frozen in liquid Nitrogen and was stored at -80°C. The protein concentration was estimated in spectrophotometer at 280nm and was analyzed along the known concentration of Bovine Serum Albumin (BSA) standards on Sodium Dodecyl-Poly Acrylamide Gel electrophoresis (SDS-PAGE).

Purification of GST-Hof1^{N (1-350)}

Pellets were kept on ice for 10 min. and was subjected to lysis with the Lysis Buffer (PBS, 0.5 mM EDTA, 1 mM DTT, 1 mg/ml lysozyme, and complete mini-EDTA-free protease inhibitor cocktail tablets). The clarified lysate was incubated with 1% Triton-X-100 for 20 min. on ice and centrifuged at 4°C, 22,000xg for 30 min. The lysate was then transferred to a tube containing glutathione sepharose-4B resin (cat. no. GE17-0756-01, GE) washed with wash buffer (5x) (PBS, 0.5 mM EDTA and 1 mM DTT) and was incubated in 4°C for 2-3 hrs. The lysate with glutathione Sepharose resins were then transferred to poly-prep chromatography columns after incubation, washed with wash buffer and was stored in 4°C.

In vitro Binding assay

GST or GST-Hof1^{N (1-350)} (~3 μg) bound to glutathione beads were mixed with ~1 μg 6His-bdSUMO-Gin4^{KA1} and incubated in binding buffer (PBS, 0.2% NP-40) for 1hour at 4°C. The beads were washed 3-5 times with wash buffer (PBS, 1% NP-40) and were analyzed by western blot using Anti-HIS (H-3) mouse monoclonal IgG1 (Santa Cruz/#sc-8036/Lot. No. C0421), Anti-GST (B-14) mouse monoclonal IgG1 (Santa Cruz/#sc-138/Lot. No. K1020) and Anti-mouse IgG-HRP (Cell Signalling/#7076S/Lot.No. 36). Bound Gin4^{KA1} was calculated as $(IGin4^{KA1} - I_{background}) / (IGST-Hof1^{N(1-350)} - I_{background})$, where I is the Integrated density value of the protein bands measured using ImageJ (NIH).

Spot assay

The yeast strains were grown in liquid YPD broth overnight at 23°C. The cultures were diluted to 1.0 O.D. and was serially diluted in the order 10^0 , 10^{-1} , 10^{-2} , 10^{-3} , and 10^{-4} . These dilutions were spotted on three YPD plates and incubated at 23°C, 30°C and 37°C. The growth was monitored, and plates were scanned after 48 hrs. of incubation.

The strains used for rescue of $\Delta gin4\Delta hof1$ synthetic lethality (**Fig. S3J** [↗](#)) were grown overnight in S.C. media lacking Uracil and Leucine. The culture was diluted to an O.D. of 1.0 and was serially diluted in the order 10^0 , 10^{-1} , 10^{-2} , 10^{-3} , and 10^{-4} . These dilutions were spotted on three SC-Ura-Leu plates and were incubated at 23°C for two days. Subsequently, the colonies were patched onto the 5-FOA plates (concentration 1mg/ml) to select for Ura-cells.

Statistical analysis

The experiments were performed three independent times. The significance values for the graphs plotted were extracted using the One-way ANOVA -Kruskal-Wallis multiple comparison test (plots with residence time), and Tukey's multiple comparison test (Bar and Stacked column plots), an inbuilt function in GraphPad Prism (v.6.04). A p-value of $p < 0.05$ was considered statistically significant. (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, ns: $p > 0.05$).

Supplemental figures and tables

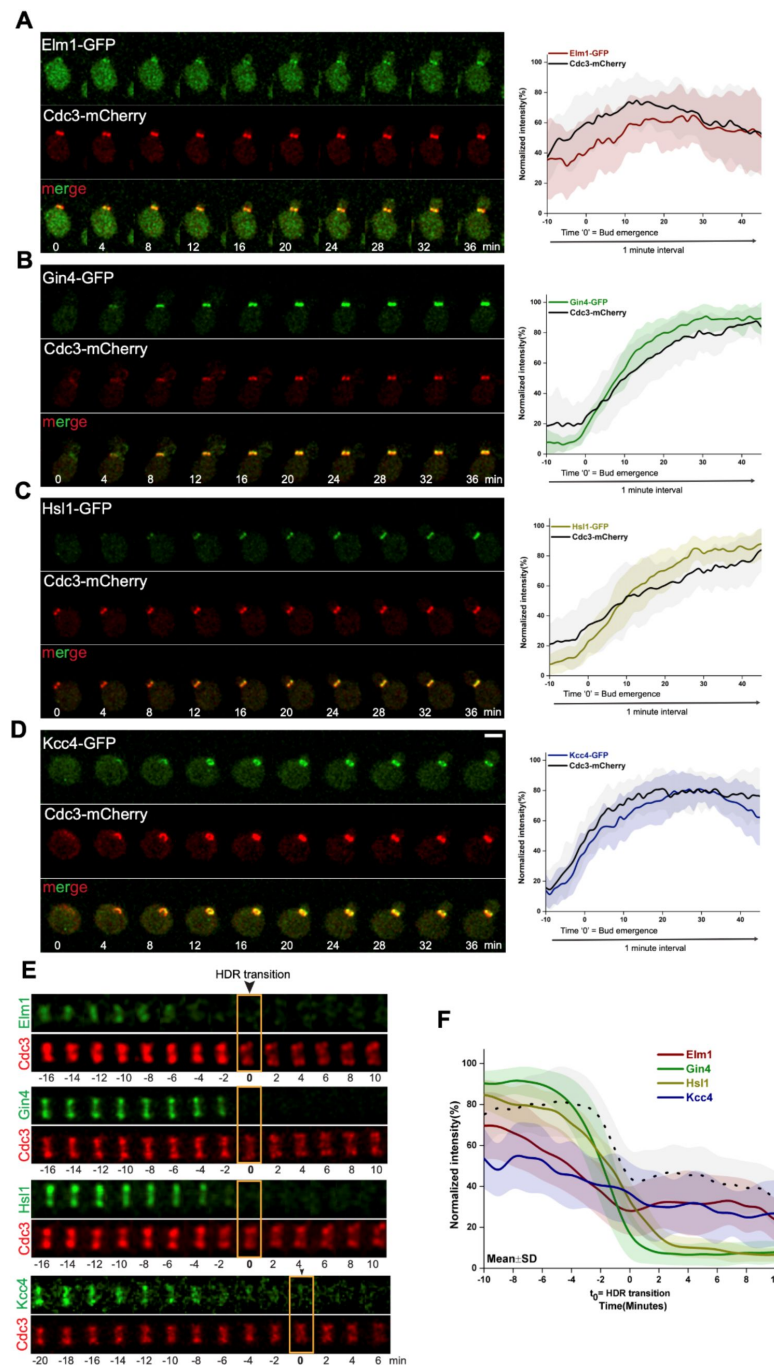


Figure S1.

Dynamics of septin-associated kinases exhibiting a sequential pattern of recruitment during bud emergence and derly disassembly during the HDR transition.

(A) Representative time-lapse images and graph for kinetics of Elm1-GFP during d-emergence ($t=0$) ($n>15$ cells). (B) Representative time-lapse images and graph for kinetics of Gin4-GFP during bud-emergence 0 ($n>15$ cells). (C) Representative time-lapse images and graph for kinetics of Hsl1-GFP during bud-emergence ($t=0$) ($n>15$ ls). (D) Representative time-lapse images and graph for kinetics of Kcc4-GFP during bud-emergence ($t=0$) ($n>15$ cells). (E) representative montages of Elm1-GFP, Gin4-GFP, Hsl1-GFP and Kcc4-GFP during HDR transition (t_0 =septin splitting). (F) A plot normalized signal intensity at bud neck for indicated GFP-tagged proteins in (E). Cdc3-mCherry profile of Elm1 is plotted as a reference for septin splitting ($n>14$ cells/strain).

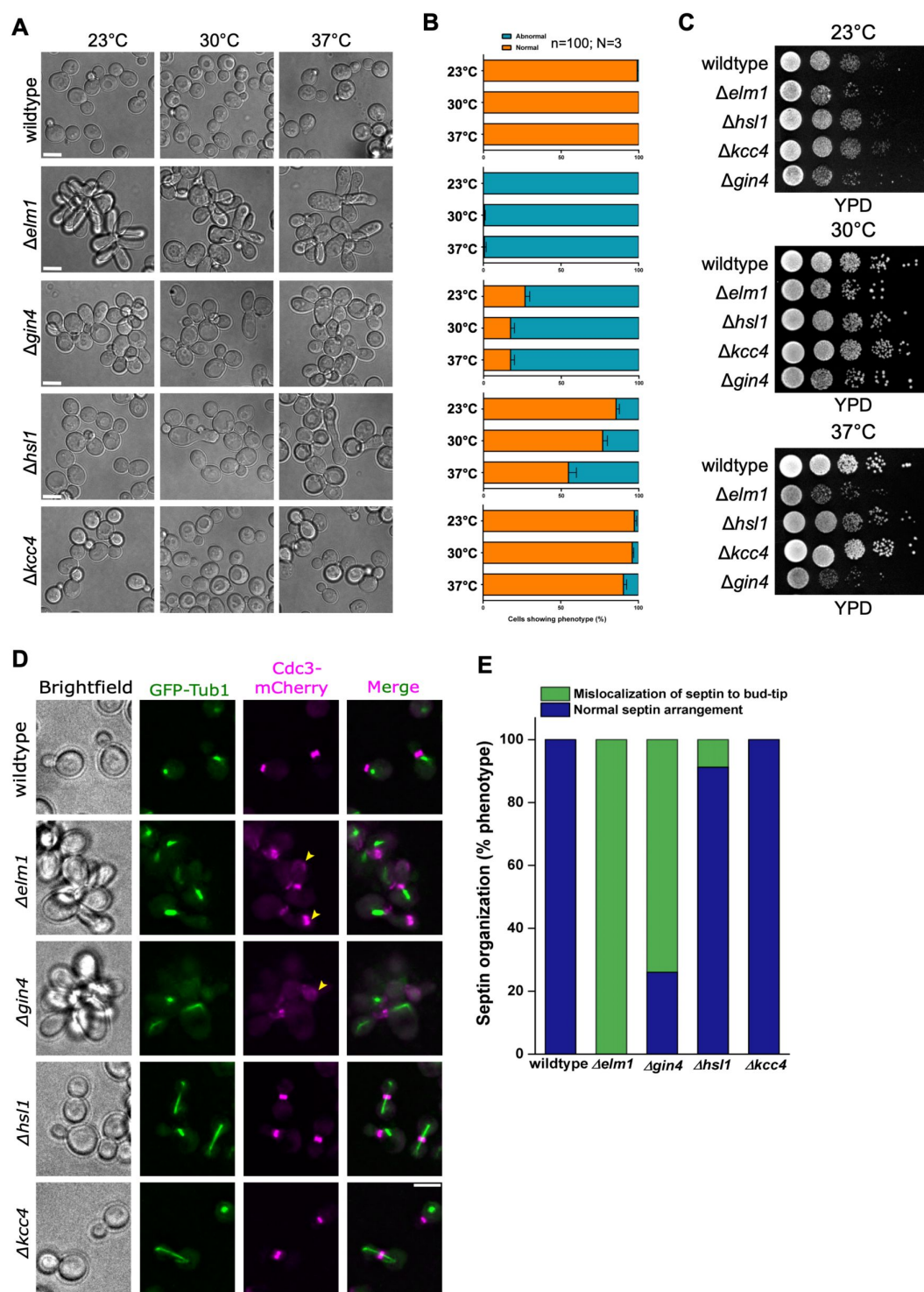


Figure S2.

Septin kinases deletion differentially perturbs septin organization and alters cell morphology.

(A) Images showing the morphological defects in $\Delta elm1$, $\Delta gin4$, $\Delta kcc4$ and $\Delta hsl1$ at 23°C, 30°C and 37°C. (B) Bar graph showing % phenotype for indicated strains of (A). (C) Growth assay for indicated strains of (A). (D) Representative images showing septin defects in kinase deletions during bud-emergence. Arrows depict mislocalised septins in $\Delta elm1$ and $\Delta gin4$ strains, scale bar=5µm. (E) The bar plot of septin mislocalisation for strains indicated in (D) (n>25 cells/strain).

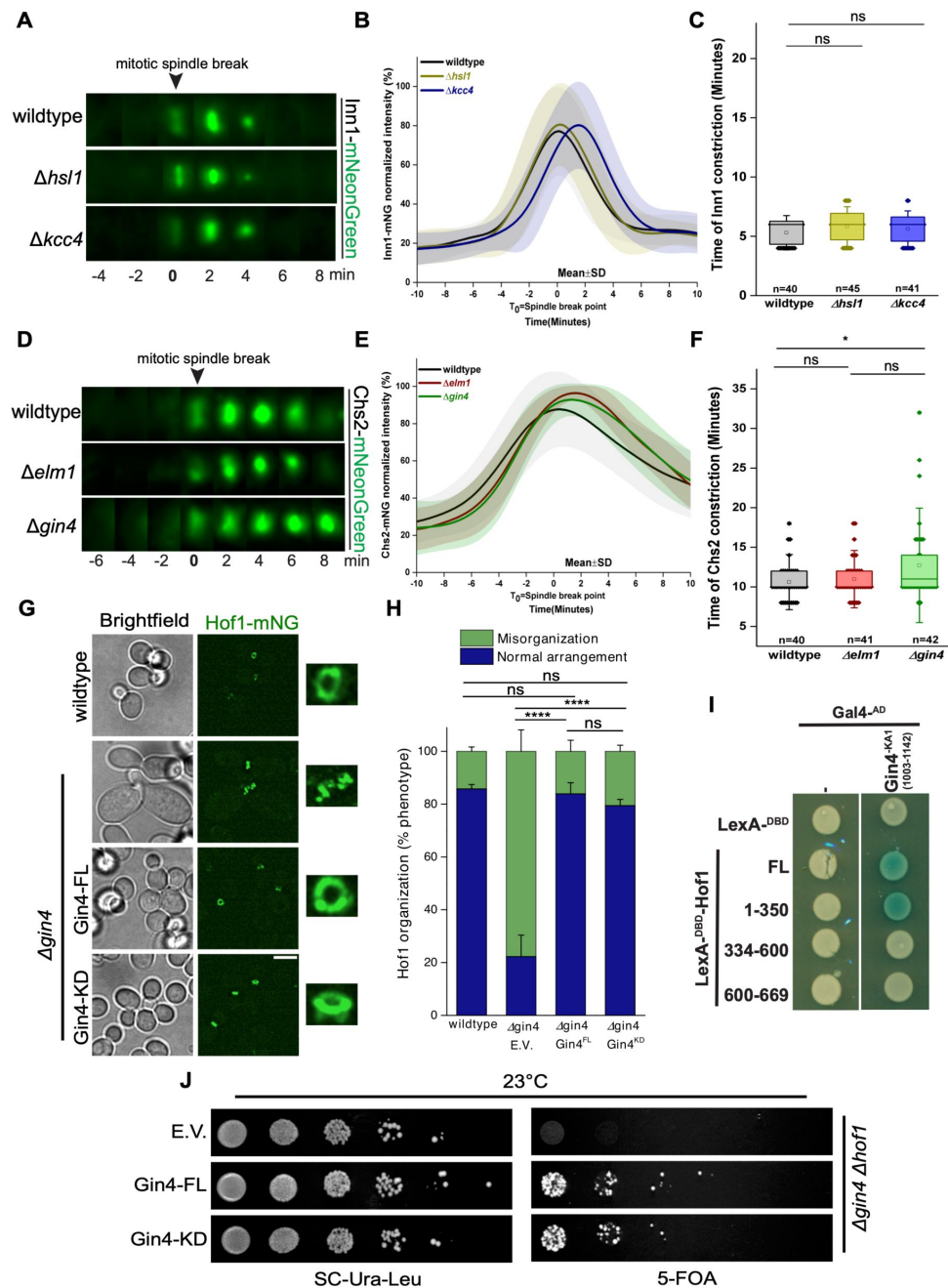


Figure S3.

Actomyosin ring dynamics is unaltered in $\Delta hsl1$ and $\Delta kcc4$ deletions, and $Gin4$ regulate primary septum (PS) ormaton in a kinase-independent manner.

(A) Representative time-lapse images of Inn1-mNG in wildtype, $\Delta hsl1$ and $\Delta kcc4$. **(B)** Plot of normalized signal intensity of Inn1-mNG in indicated strains of (A) (t_0 =spindle breakpoint). **(C)** Quantification for Inn1-residence in indicated strains of (A). ns: $p > 0.05$. **(D)** Representative time-lapse images of Chs2-mNG in wildtype, $\Delta elm1$ and $\Delta gin4$. **(E)** Plot of normalized signal intensity of Chs2-mNG in indicated strains of (D); a population of cells showing asymmetric constriction in $\Delta elm1$ and $\Delta gin4$ is plotted (t_0 =spindle breakpoint). **(F)** Quantitative analysis of Chs2-residence in indicated strains of (D). *: $p < 0.05$, ns: $p > 0.05$. **(G)** Representative images for the rescue of Hof1-misorganization by $Gin4^{FL}$ and $Gin4^{KD}$ in $\Delta gin4$ cells. Scale bar = 5 μ m. **(H)** Quantification for indicated strains of (G), (N=3 and $n > 130$ cells/strain). ****: $p < 0.0001$, ns: $p > 0.05$ corresponds to population of cells with disorganized Hof1 at the bud neck. **(I)** Representative images of Yeast Two-Hybrid for interaction between in4-KA1 domain and different domains of Hof1. **(J)** Spot assay for the rescue of synthetic lethality by $Gin4^{FL}$ and $Gin4^{KD}$ in $\Delta gin4$ $hof1$ cells at 23°C.

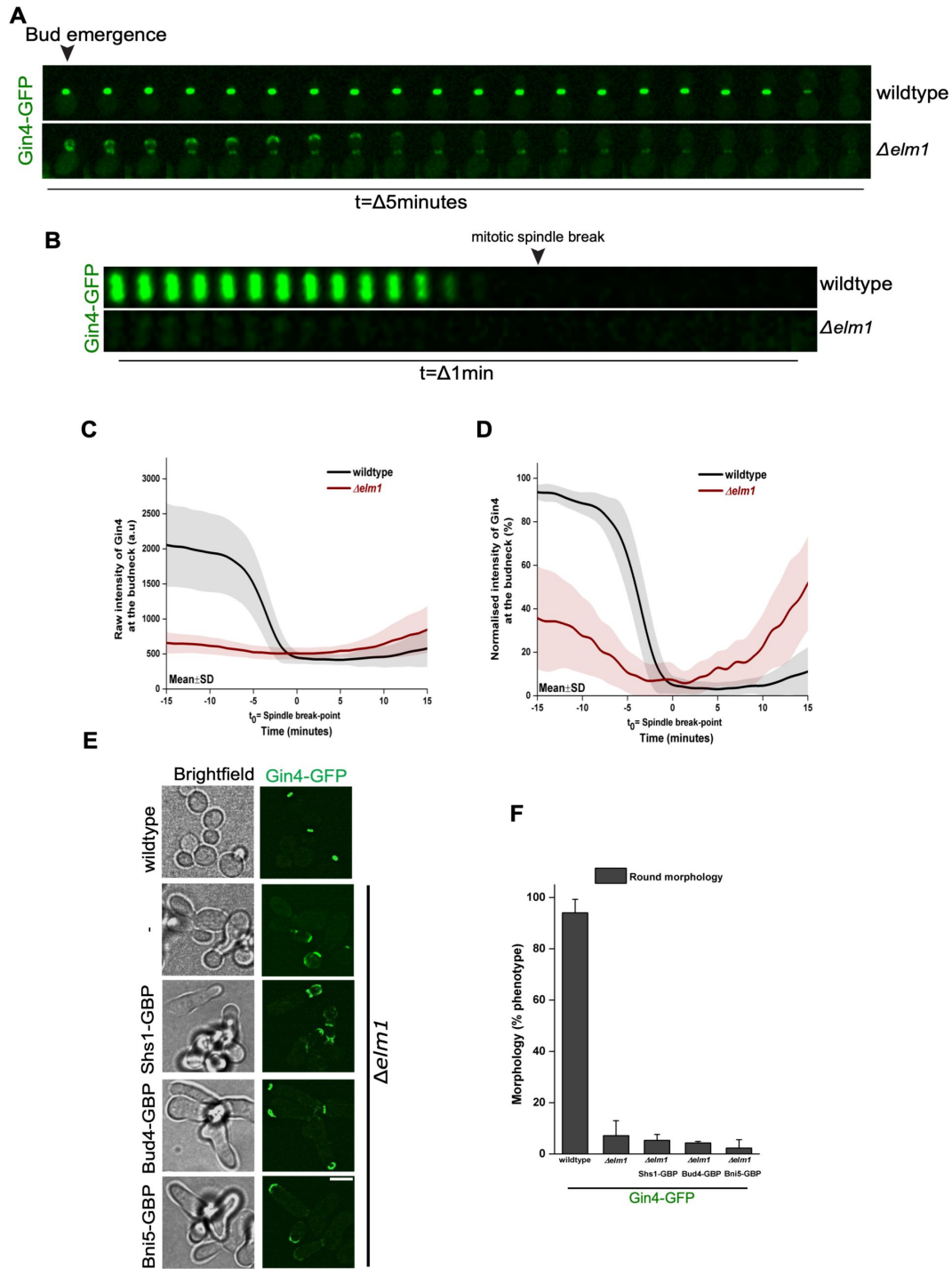


Figure S4.

A GBP-based tethering screen for targeted localization of Gin4-GFP to the bud neck in $\Delta elm1$ via different bud neck proteins.

(A) Representative montages for recruitment of Gin4-GFP during bud emergence in $\Delta elm1$ background. (B) Representative montages of the bud neck depicting the localisation of Gin4 in $\Delta elm1$. (C) Plot of raw intensity of Gin4-GFP in indicated strains of (B) (t_0 =spindle breakpoint). (D) Plot of normalized signal intensity of Gin4-GFP in indicated strains of (B) (t_0 =spindle breakpoint). (E) Representative images showing artificial tethering of Gin4-GFP via Shs1-GBP, Bud4-GBP and Bni5-GBP in $\Delta elm1$. Scale bar=5 μ m. (F) Bar graph showing % phenotype for indicated strains of (E) (N=3 and n>350/strain).

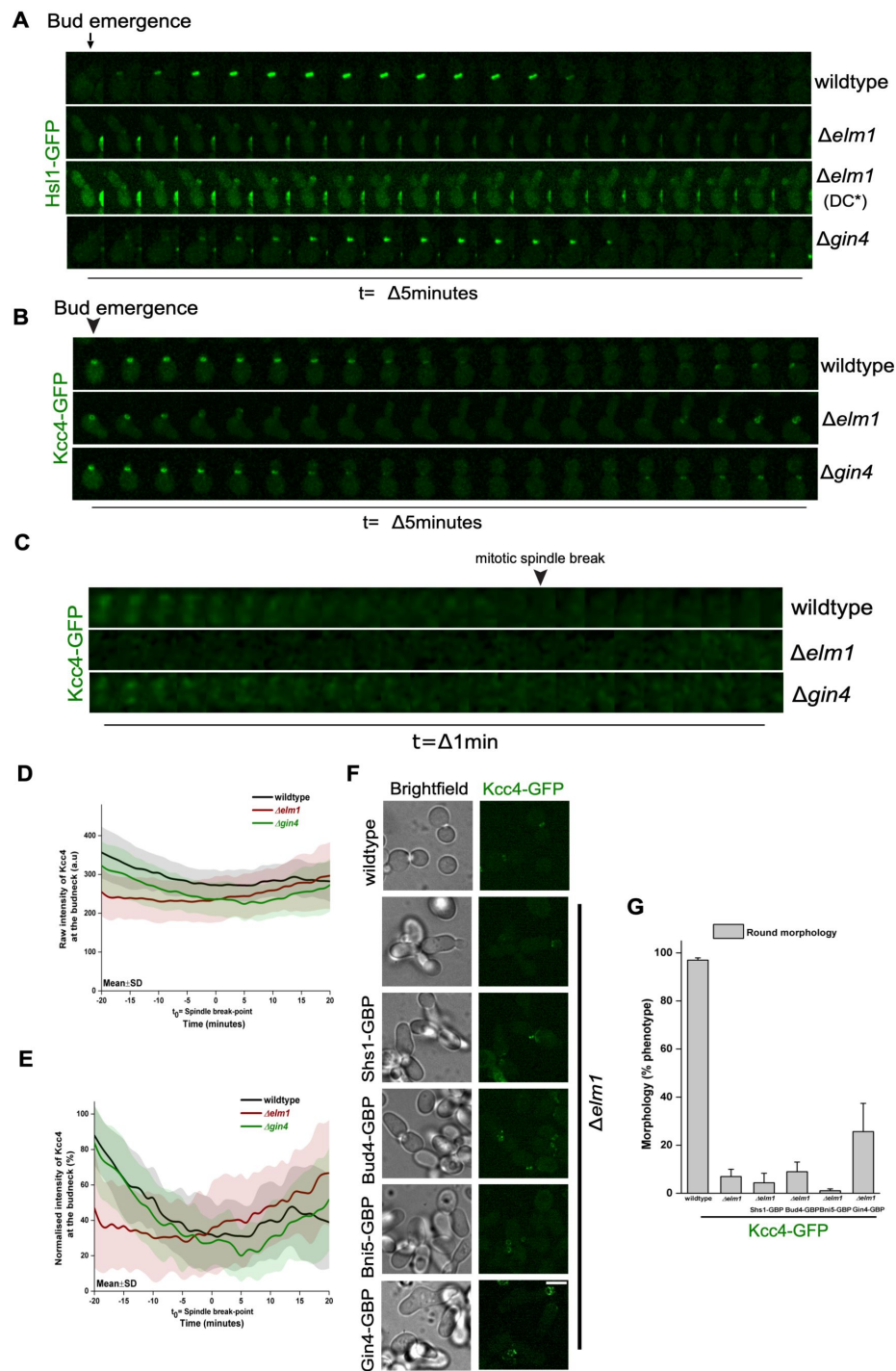


Figure S5.

A GBP-based screen for artificially restoring Kcc4 localisation in $\Delta elm1$ cells.

(A) Representative montages for recruitment of Hsl1-GFP during bud emergence in $\Delta elm1$ and $\Delta gin4$ background (DC* indicates Differential Contrast). (B) Representative montages for recruitment of Kcc4-GFP during bud emergence in $\Delta elm1$ and $\Delta gin4$ backgrounds. (C) Representative montages of the bud neck depicting the localisation of Kcc4 in $\Delta elm1$ and $\Delta gin4$ cells. (D) Plot of raw intensity of Kcc4-GFP in indicated strains of (C) (t_0 =spindle breakpoint). (E) Plot of normalized signal intensity of Kcc4-GFP in indicated strains of (C) (t_0 =spindle breakpoint). (F) Representative images showing artificial tethering of Kcc4-GFP with Shs1-GBP, Bud4-GBP, Bni5-GBP and Gin4-GBP in $\Delta elm1$. Scale bar=5 μ m. (G) Bar graph showing % phenotype for indicated strains of (F) (N=3, n>350 cells/strain).

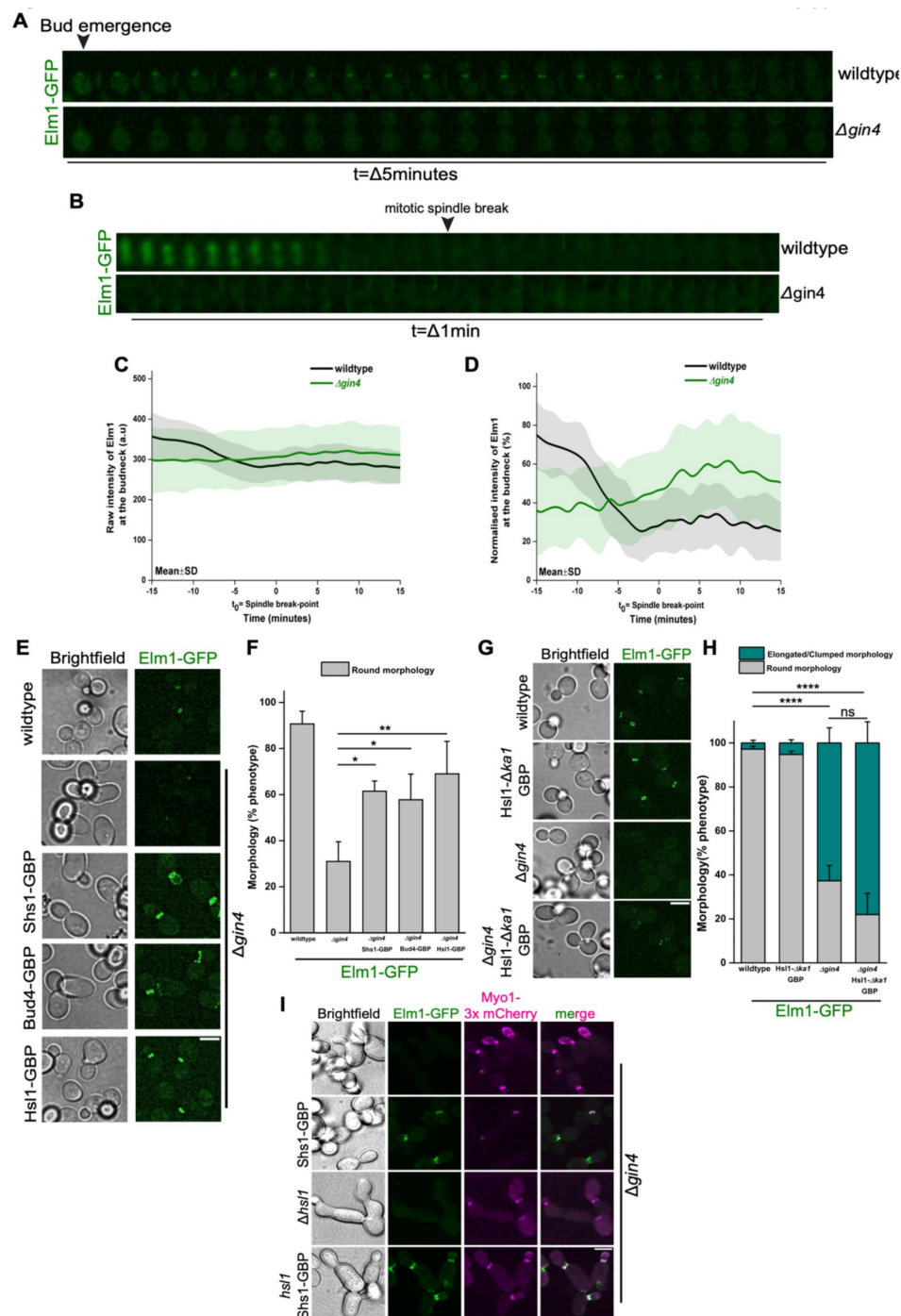


Figure S6.

Hsl1 kinase becomes essential downstream of Elm1 to regulate cytokinesis in $\Delta gin4$.

(A) Representative montages for recruitment of Elm1-GFP during bud emergence in $\Delta gin4$ background. (B) Representative montages of the bud neck depicting the localisation of Elm1 in $\Delta gin4$. (C) Plot of raw intensity of Elm1-GFP in indicated strains of (B) (t_0 =spindle breakpoint). (D) Plot of normalized signal intensity of Elm1-GFP in indicated strains of (B) (t_0 =spindle breakpoint). (E) Representative images showing artificial tethering of Elm1-GFP with Shs1-GBP, Bud4-GBP and Hsl1-GBP in $\Delta gin4$. Scale bar=5 μm . (F) Bar graph showing % phenotype for indicated strains of (E) (N=3 and n>350/strain). *: p<0.05, **: p<0.01 (G) Representative images showing tethering of Elm1-GFP with Hsl1- $\Delta ka1$ -GBP in $\Delta gin4$. Scale bar=5 μm . (H) Bar graph showing % phenotype for indicated strains of (G) (N=3, n>300 cells/strain). ****: p<0.0001, ns: p>0.05 corresponds to population of cells with round morphology. (I) Representative images showing localisation of Myo1-3xmCherry in the $\Delta gin4$ $\Delta hsl1$ cells.

Strain number	Genotype	Source
YSP002	<i>MATa ura3-52 leu2Δ1 trp1Δ63 his3Δ200 (ESM356)</i>	100
YSP120	<i>ESM356-Elm1-GFP-Hph Cdc3-mCherry-Trp</i>	This study
YSP121	<i>ESM356-Gin4-GFP-Hph Cdc3-mCherry-Trp</i>	This study
YSP122	<i>ESM356-Kcc4-GFP-Hph Cdc3-mCherry-Trp</i>	This study
YSP123	<i>ESM356-Hsl1-GFP-Hph Cdc3-mCherry-Trp</i>	This study
YSP524	<i>ESM356-Cdc3-mCherry-Trp, Tub1-GFP-ura</i>	This study
YSP520	<i>ESM356-Cdc3-mCherry-Trp, Tub1-GFP-Ura, Δelm1::His3mX6</i>	This study
YSP521	<i>ESM356-Cdc3-mCherry-Trp, Tub1-GFP-Ura, Δgin4::His3mX6</i>	This study
YSP1104	<i>ESM356-Cdc3-mCherry-Trp, Tub1-GFP-Ura, Δkcc4::His3mX6</i>	This study
YSP518	<i>ESM356-Δhsl1-His3mX6, Cdc3-mCherry-Trp, Tub1-GFP-Ura</i>	This study
YSP058	<i>ESM356-Δelm1::His3MX6</i>	This study
YSP062	<i>ESM356-Δgin4::His3MX6</i>	This study
YSP050	<i>ESM356-Δhsl1::His3MX6</i>	This study
YSP115	<i>ESM356-Δkcc4::KanMX6</i>	This study
YSP950	<i>ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1017	<i>ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δelm1::3HA-Trp</i>	This study
YSP1004	<i>ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δgin4::3HA-Trp</i>	This study
YSP1014	<i>ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δhsl1::3HA-Trp</i>	This study
YSP1037	<i>ESM356-Inn1-mNG-Kanmx6, pHis3-mRuby2-Tub1-Ura3, Δkcc4::3HA-Trp</i>	This study
YSP948	<i>ESM356-Hof1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1024	<i>ESM356-Hof1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δelm1::3HA-Trp</i>	This study
YSP1095	<i>ESM356-Hof1-mNG-Kanmx6, pHis3-mRuby2-Tub1-Ura3, Δgin4::3HA-Trp</i>	This study
YSP947	<i>ESM356-Chs2-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1019	<i>ESM356-Chs2-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δelm1::3HA-Trp</i>	This study
YSP1002	<i>ESM356-Chs2-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δgin4::3HA-Trp</i>	This study
YSP003	<i>MATa ura3-52 lys2-801amber ade2-101ochre trp1Δ63 his3Δ200 leu2Δ1 (YPH500)</i>	94
YSP004	<i>MATa leu2 ADE2 ura3-52::URA3-lexA-op-LacZ his3 trp (SGY37)</i>	94
YSP997	<i>ESM356-Cdc3-mCherry-Ura, Δgin4::3HA-Trp</i>	This study
YSP1581	<i>ESM356-Cdc3-mCherry-Ura Δgin4::3HA-Trp pRS305-(Vector alone) cut with Kas I and integrated into Leu locus</i>	This study
YSP1582	<i>ESM356-Cdc3-mCherry-Ura Δgin4::3HA-Trp pRS305-pTEF-Gin4(FL)-GFP-tCyc1 cut with Kas I and integrated into Leu locus</i>	This study

Supplemental Table 1

Yeast strains used in this study

YSP1583	ESM356-Cdc3-mCherry-Ura Δ gin4::3HA-Trp pRS305-pTEF-Gin4(K48A)-GFP-tCyc1 cut with Kas I and integrated into Leu locus	This study
YSP1236	ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δ gin4::3HA-Trp, pRS305-(Empty vector)-cut with Kas I and integrated into leu locus	This study
YSP1237	ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δ gin4::3HA-Trp, pRS305-pGin4-Gin4(FL)-tGin4-cut with Kas I and integrated into leu locus	This study
YSP1238	ESM356-Inn1-mNG-KanMX6, pHis3-mRuby2-Tub1-Ura3, Δ gin4::3HA-Trp, pRS305-pGin4-Gin4(Kinase dead)-tGin4-cut with Kas I and integrated into leu locus	This study
YSP1312	ESM356-Hof1-mNG-Kanmx6, pHis3-mRuby2-Tub1-Ura3, Δ gin4::3HA-Trp, pRS305-(Empty vector)-cut with Kas1 and integrated into leu locus	This study
YSP1313	ESM356-Hof1-mNG-Kanmx6, pHis3-mRuby2-Tub1-Ura3, Δ gin4::3HA-Trp, pRS305-pGin4-Gin4(FL)-tGin4-cut with Kas1 and integrated into leu locus	This study
YSP1314	ESM356-Hof1-mNG-Kanmx6, pHis3-mRuby2-Tub1-Ura3, Δ gin4::3HA-Trp, pRS305-pGin4-Gin4(Kinase dead)-tGin4-cut with Kas1 and integrated into leu locus	This study
YSP001	MATa ura3-52 lys2-801amber ade2-101ochre trp1 Δ 63 his3 Δ 200 leu2 Δ 1 (YPH499)	94
YSP1086	YPH499- Δ hof1::TRP1 pRS316-HOF1 Δ gin4::HIS3MX6 (MFY1474)	71
YSP1357	YPH499 Δ hof1::TRP1 pRS316-HOF1 Δ gin4::HIS3MX6, pRS305-Empty Vector-cut with Kas1 and integrated into Leu locus	This study
YSP1358	YPH499 Δ hof1::TRP1 pRS316-HOF1 Δ gin4::HIS3MX6, pRS305-pGin4-Gin4(FL)-tGin4-cut with Kas1 and integrated into Leu locus	This study
YSP1359	YPH499 Δ hof1::TRP1 pRS316-HOF1 Δ gin4::HIS3MX6, pRS305-pGin4-Gin4(K48A)-tGin4-cut with Kas1 and integrated into Leu locus	This study
YSP034	ESM356-Gin4-GFP-Hph	This study
YSP1345	ESM356-GFP-Gin4-GFP-Hph Δ elm1::3HA-Trp	This study
YSP1419	ESM356-Gin4-GFP-Hph, Δ elm1::3HA-Trp, Hsl1-GBP-KanMX6	This study
YSP1651	ESM356- Δ elm1::3HA-Trp Δ gin4::6HA NatNT2	This study
YSP1665	ESM356- Δ elm1::3HA-Trp Δ gin4::6HA NatNT2 pRS305 cut with Kas I and integrated into Leu locus	This study
YSP1666	ESM356- Δ elm1::3HA-Trp Δ gin4::6HA NatNT2 pRS305-pTEF-Gin4(FL)-GFP-tCyc1 cut with Kas I and integrated into Leu locus	This study
YSP1667	ESM356- Δ elm1::3HA-Trp Δ gin4::6HA NatNT2 pRS305-pTEF-Gin4(K48A)-GFP-tCyc1 cut with Kas I and integrated into Leu locus	This study
YSP1658	ESM356- Δ elm1::3HA-Trp Δ gin4::6HA NatNT2 pRS305 cut with Kas I and integrated into Leu locus Hsl1-GBP-KanMX6	This study
YSP1652	ESM356- Δ elm1::3HA-Trp Δ gin4::6HA NatNT2 pRS305-pTEF-Gin4(FL)-GFP-tCyc1 cut with Kas I and integrated into Leu locus Hsl1-GBP-KanMX6	This study

Supplemental Table 1 (continued)

YSP1653	<i>ESM356-Δelm1::3HA-Trp Δgin4::6HA NatNT2 pRS305-pTEF-Gin4(K48A)-GFP-tCyc1 cut with Kas I and integrated into Leu locus Hsl1-GBP-KanMX6</i>	This study
YSP1737	<i>ESM356-Gin4-GFP-Hph, Hsl1-Δka1-GBP-KanMX6</i>	This study
YSP1738	<i>ESM356-Gin4-GFP-Hph Δelm1::3HA-Trp Hsl1-Δka1-GBP-KanMX6</i>	This study
YSP1590	<i>ESM356-Gin4-GFP-Hph mRuby-Tub1-Ura</i>	This study
YSP1594	<i>ESM356-Gin4-GFP-Hph Δelm1::3HA-Trp, pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1411	<i>ESM356-Gin4-GFP-Hph, Δelm1::3HA-Trp Shs1-GBP-KanMX6</i>	This study
YSP1415	<i>ESM356-Gin4-GFP-Hph, Δelm1::3HA-Trp, Bud4-GBP-KanMX6</i>	This study
YSP1423	<i>ESM356-Gin4-GFP-Hph, Δelm1::3HA-Trp, Bni5-GBP-KanMX6</i>	This study
YSP119	<i>ESM356-Hsl1-GFP-Hph</i>	This study
YSP1485	<i>ESM356-Hsl1-GFP-Hph Δelm1::3HA-Trp</i>	This study
YSP1605	<i>ESM356-Hsl1-GFP-Hph Δelm1::3HA-Trp Shs1-GBP-KanMX6</i>	This study
YSP1606	<i>ESM356-Hsl1-GFP-Hph Δelm1::3HA-Trp Bud4-GBP-KanMX6</i>	This study
YSP1740	<i>ESM356-Hsl1-GFP-Hph, Δelm1::3HA-Trp Kcc4-GBP-KanMX6</i>	This study
YSP1745	<i>ESM356-Hsl1-GFP-Hph, Δelm1::3HA-Trp Bni5-GBP-KanMX6</i>	This study
YSP1560	<i>ESM356-Hsl1-GFP-Hph, Δelm1::3HA-Trp Gin4-GBP-KanMX6</i>	This study
YSP036	<i>ESM356-Kcc4-GFP-Hph</i>	This study
YSP1482	<i>ESM356-Kcc4-GFP-Hph Δelm1::3HA-Trp</i>	This study
YSP1734	<i>ESM356-Kcc4-GFP-Hph, Δelm1::3HA-Trp Hsl1-GBP-KanMX6</i>	This study
YSP1730	<i>ESM356-Kcc4-GFP-Hph, Δelm1::3HA-Trp Shs1-GBP-KanMX6</i>	This study
YSP1732	<i>ESM356-Kcc4-GFP-Hph, Δelm1::3HA-Trp Bud4-GBP-KanMX6</i>	This study
YSP1747	<i>ESM356-Kcc4-GFP-Hph, Δelm1::3HA-Trp Bni5-GBP-KanMX6</i>	This study
YSP1518	<i>ESM356-Kcc4-GFP-Hph Δelm1::3HA-Trp Gin4-GBP-KanMX6</i>	This study
YSP1906	<i>ESM356-Kcc4-GFP-Hph pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1908	<i>ESM356-Hsl1-GFP-Hph pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1910	<i>ESM356-Kcc4-GFP-Hph Δelm1::3HA-Trp pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1912	<i>ESM356-Kcc4-GFP-Hph Δgin4::3HA-Trp pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1914	<i>ESM356-Hsl1-GFP-Hph Δelm1::3HA-Trp pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1916	<i>ESM356-Hsl1-GFP-Hph Δgin4::3HA-Trp pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1588	<i>ESM356-Elm1-GFP-Hph pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP1592	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-Trp pHis3-mRuby2-Tub1-Ura3</i>	This study
YSP032	<i>ESM356-Elm1-GFP-Hph</i>	This study
YSP1346	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-Trp</i>	This study
YSP1410	<i>ESM356-Elm1-GFP-Hph, Δgin4::3HA-Trp Shs1-GBP-KanMX6</i>	This study
YSP1414	<i>ESM356-Elm1-GFP-Hph, Δgin4::3HA-Trp, Bud4-GBP-KanMX6</i>	This study
YSP1418	<i>ESM356-Elm1-GFP-Hph, Δgin4::3HA-Trp, Hsl1-GBP-KanMX6</i>	This study
YSP1422	<i>ESM356-Elm1-GFP-Hph, Δgin4::3HA-Trp, Bni5-GBP-KanMX6</i>	This study

Supplemental Table 1 (continued)

YSP1735	<i>ESM356-Elm1-GFP-Hph, Hsl1-Δka1-GBP-KanMX6</i>	This study
YSP1736	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-Trp Hsl1-Δka1-GBP-KanMX6</i>	This study
YSP1591	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-Trp Cdc3-mCherry-Ura</i>	This study
YSP1453	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-Trp, Shs1-GBP-KanMX6, Cdc3-mCherry-Ura</i>	This study
YSP1715	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Δhsl1::natNT1 Cdc3-mCherry-Ura</i>	This study
YSP1717	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Shs1-GBP-KanMX6 Δhsl1::natNT1 Cdc3-mCherry-Ura</i>	This study
YSP1757	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Inn1-3xmCherry-His3MX6</i>	This study
YSP1759	<i>ESM356-Elm1-GFP-Hph, Δgin4::3HA-Trp Shs1-GBP-KanMX6 Inn1-3xmCherry-His3MX6</i>	This study
YSP1755	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Δhsl1::natNT1 Inn1-3xmCherry-His3MX6</i>	This study
YSP1754	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Shs1-GBP-KanMX6 Δhsl1::natNT1 Inn1-3xmCherry-His3MX6</i>	This study
YSP1724	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Myo1-3xmCherry-His3MX6</i>	This study
YSP1726	<i>ESM356-Elm1-GFP-Hph, Δgin4::3HA-Trp Shs1-GBP-KanMX6 Myo1-3xmCherry-His3MX6</i>	This study
YSP1722	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Δhsl1::natNT1 Myo1-3xmCherry-His3MX6</i>	This study
YSP1721	<i>ESM356-Elm1-GFP-Hph Δgin4::3HA-trp Shs1-GBP-KanMX6 Δhsl1::natNT1 Myo1-3xmCherry-His3MX6</i>	This study

Supplemental Table 1 (continued)

Primer number	Primer name	Sequence
P86	Elm1-S1	GATATTATTTTTGAACGCCAGGTAAACAATAATTACTT AGCATGAAATGCGTACGCTGCAGGTCGAC
P87	Elm1-S2	GATATTATTTTTGAACGCCAGGTAAACAATAATTACTT AGCATGAAATGCGTACGCTGCAGGTCGACTCCGACAG ATATCATCCTGTAGTTTCATCTAATCGATGAATTCGAGC TCG
P88	Elm1-S3	CAGTAGAAGATGTAAAAACGTATCTGAACCTTGCAGAT AATGGTCAAATACGTACGCTGCAGGTCGAC
P110	Gin4-S1	TTGATCAAGAAAGATATTCGCAGCACAATAACAATAATA CATTCAAAATGCGTACGCTGCAGGTCGAC
P111	Gin4-S2	GGCAGAACAAAACGAAGGAGACAAAACATGATTGCATT ACATTAGCACTAATCGATGAATTCGAGCTCG
P112	Gin4-S3	TGGTCAATGAAGTTGAGAACGTCCTGAATAAGGAAGGC GTTCTACAAAACGTACGCTGCAGGTCGAC
P113	Hsl1-S1	ACTCCTTTATTATTTGTTATTAATATTTTTATTTTTACCA CACGACATGCGTACGCTGCAGGTCGAC
P114	Hsl1-S2	AATTTTTCAAATTATTGTTGTATAATTATATAACATCTAT ATAGAATTTAATCGATGAATTCGAGCTCG
P115	Hsl1-S3	AAAAGTTTAATGATGATGTGGAGAGAGTAATTCGAAAT GCCGGACGTTACGTACGCTGCAGGTCGAC

Supplemental Table 2

Oligonucleotides used in this study

P107	Kcc4-S1	ATAGGCTTGATAATAGTGCTGACGTAAATTACCAGAAC TACTGCAGTATGCGTACGCTGCAGGTCGAC
P108	Kcc4-S2	GTTACATCGTATTGTCCATTTGGGGATCGATTATCCCTC CCTTTTTTTTAATCGATGAATTCGAGCTCG
P109	Kcc4-S3	TTGCTACAGAAATCCAAATTATTTTACAAAAAGAAGGTG TTTTGGACAAACGTACGCTGCAGGTCGAC
P90	Hof1-S2	TTTCTTTTATCAGAAAAGTAGTAAATTTGATATACATCGA GATCAatcgatgaattcgagctcg
P91	Hof1-S3	GGATTAATTCCCTATAATTTTATTTCAGCTACTGCATCAA GGTCTTcgtagctgcaggctcgac
P698	Inn1-S2	TTCTATTTAAATATTAGTATTACAATAACAGCATGTTTCT CTTGTCATTAATCGATGAATTCGAGCTCG
P699	Inn1-S3	GGAAGAATTCGATGTCTCCTACAAGAAAAAGACCACCT CCAAGGCTCAGCCGTACGCTGCAGGTCGAC
P742	Chs2-S2	TGAGAAAAAGAGGGAATGACGAGAAATTAGCTGAAAA ATACTGGCATTAAATCGATGAATTCGAGCTCG
P741	Chs2-S3	CATGGGAAGTCTCTAAATTAGACTTACCAAATGTTTTCC ACAAAAAGGCCGTACGCTGCAGGTCGAC
P855	pRS305-Gin4- Fwd	GGCGGCCGCTCTAGAACTAGTCTGGTGTTAAAGCAGA AATCACG
P1021	pRS305-Gin4-Rvs	atcgaattcctgcagcccgCGGTTATACAAATAGTAATCAACA CG
P856	Gin4-K48A-Rvs	TGCTTTTGATATTACAGCAACTGCCGCCTCTTGT
P857	Gin4-K48A-Fwd	CAAGAGGCGGCAGTTGCTGTAAATCAAAAGCAG
P60	Shs1-S2	TATTTATTTATTTATTTGCTCAGCTTTGGATTTTGTACAG ATACAACTCAATCGATGAATTCGAGCTCG
P61	Shs1-S3	AAAAAAATGACACGTATACTGATTTAGCCTCTATTGCAT CGGGTAGAGATCGTACGCTGCAGGTCGAC
P81	Bud4-S2	CTCTCAGTCATCAGATTATATGCTGTTTTTCATTCCATTA ATCACCTTCTAATCGATGAATTCGAGCTCG
P82	Bud4-S3	AGGAGGAAAAGACTCGAACAACCGGTCATAATTTGAAG CAAGATTTTAATCGTACGCTGCAGGTCGAC
P84	Bni5-S2	TCTGTCTTAATTTATAAATATTTATAACAACCCTTGCGT AATGTAATCAATCGATGAATTCGAGCTCG
P85	Bni5-S3	GAAAGCAAATTGATCGTGGCGAAAATGGCCAATTTTGG ATTGGAATAAACGTACGCTGCAGGTCGAC
P906	Hof1-(1-350 aa)- Fwd	ATCTGATCGAAGGTCGTGGGATCCCATGAGCTACAGT TATGAAGCTTGTTT
P908	Hof1-350-Rvs	TGCGGCCGCTCGAGTCGACCCGGTGTTGAATTTCTAA ATTGTGTAGCCATG
P1251	pET28a- bdSUMO-Gin4- FL-Rvs	CGAGTGCGGCCGCAAGCTTGCTATTTTTGTAGAACGCC TTCCTTATTCAG
P1254	pET28a- bdSUMO- PIP2BD/KA1-Fwd	gcccaTGCTGCACCAAAGTGGAGGCGGCTCGCAGGCAAA GGAT
P340	Gin4-Fwd-Y2H	gggtctgccccggcagggaATGGCAATCAATGGTAACAGTA
P341	Gin4-Rvs-Y2H	gatatcgaattcctgcagcccCTATTTTTGTAGAACGCCTTCC

Supplemental Table 2 (continued)

P613	Gin4-KID-290-Fwd	ggttctggccccggtcagggaGGATCCCAAAAATATCCAAGTAT AAGAG
P1291	Gin4-PIP2BD/KA1(1003)-Fwd	ggttctggccccggtcagggaGGATCCACCTGTGTCAAGGCTTG ATC
P338	Elm1-Fwd	ggttctggccccggtcagggaATGTCACCTCGACAGCTTATAC
P339	Elm1-Rvs	gatatcgaattcctgcagcccCTATATTTGACCATTATCTGCA
P611	Elm1-KID-420Fwd	ggttctggccccggtcagggaGGATCCTCATCGTCCAGTGTGAA CCC
P132	Hof1-Y2H-Fwd	ggttctggccccggtcagggaATGAGCTACAGTTATGAAGCT
P133	Hof1-Y2H-Rvs	gatatcgaattcctgcagcccTCAAAGACCTTGATGCAGTAG
P618	Hof1-261-Rvs	gatatcgaattcctgcagcccTTCATCCGCTGCCGTGAATG
P619	Hof1-350-Rvs	gatatcgaattcctgcagcccTGTGGAATTTCTAAATTGTGTAG
P620	Hof1-334-Fwd	ggttctggccccggtcagggaGGATCCCCATTGCCATCTCCAGA AGT
P621	Hof1-600-Rvs	gatatcgaattcctgcagcccGACTGGAAAACCTTCACTGGTG
P622	Hof1-600-Fwd	ggttctggccccggtcagggaGGATCCGTCATTGAATATGCCAA GGC

Supplemental Table 2 (continued)

Plasmid number	Description	Source
piSP23	pFA6a-HIS3-MX6	90
piSP14	pYM25 (yeGFP-hphNT1)	89
piSP20	pYM22 (3HA-kiTRP1)	89
piSP185	pYM17 (6HA-natNT2)	90
piSP659	pDK88 (mNeongreen cassette. mNG-kanmx6)	Gift from Prof. Gislene Pereira
piSP1075	pHA29-GBP-KAN	101
piSP31	YIP211-cdc3-mCherry (URA3)	77
piSP34	YIP204-cdc3-mCherry (TRP1)	91
piSP678	pAFS125-GFP-TUB1	93
piSP1224	pHIS3p:mRuby2-Tub1+3"UTR::Ura3	92
piSP1	pRS305	94
piSP1341	pRS305-pGin4-Gin4(FL)-tGin4	This Study ^a
piSP1739	pRS305-pGin4-Gin4(K48A)-tGin4	This Study ^a
piSP1614	pRS305-pTEF-Gin4-GFP-tCYC1	This Study ^b
piSP1704	pRS305-pTEF-Gin4 (K48A)-GFP-tCYC1	This Study ^b
piSP656	pBPE39-1(3xmCherry-His3MX6 cassette)	Gift from Prof. Gislene Pereira
piSP112	pMM5S-Y2H (Empty vector)	102
piSP113	pMM6S-Y2H (Empty vector)	102
piSP635	pMM5S-Gin4-FL (1-1142aa)	This Study
piSP645	pMM6S-Gin4-FL (1-1142aa)	This Study
piSP632	pMM5S-Elm1-FL (1-640aa)	This Study

Supplemental Table 3

Plasmids used in this study

piSP642	pMM6S-Elm1-FL (1-640aa)	This Study
piSP685	pMM5S-Gin4-KID (290-1142aa)	This Study
piSP691	pMM6S-Gin4-KID (290-1142aa)	This Study
piSP683	pMM5S-Elm1-KID (421-640aa)	This Study
piSP689	pMM6S-Elm1-KID (421-640aa)	This Study
piSP568	pMM5S-Hof1-FL (1-669aa)	This Study
piSP580	pMM6S-Hof1-FL (1-669aa)	This Study
piSP1275	pMM5S-Hof1(1-350aa)	This Study
piSP1277	pMM6S-Hof1(1-350aa)	This Study
piSP1279	pMM5S-Hof1(334-600aa)	This Study
piSP1281	pMM6S-Hof1(334-600aa)	This Study
piSP1283	pMM5S-Hof1(600-669aa)	This Study
piSP1285	pMM6S-Hof1(600-669aa)	This Study
piSP1839	pMM5S-Gin4-KA1(1003-1142aa)	This Study
piSP1844	pMM6S-Gin4-KA1(1003-1142aa)	This Study
piSP65	pET28a-6HIS-bdSUMO	¹⁰³
piSP1094	pGEX-5X1	GE Healthcare, ¹⁰⁴
piSP1320	pGEX-5X1-Hof1(1-350)	This Study
piSP1712	pET28a-6HIS-bdSUMO-Gin4(KA1)	This Study

^a & ^b plasmids piSP1341, piSP1739, piSP1614 and piSP1704 were cut with Kas I restriction enzyme and integrated into *Leu* locus.

Supplemental Table 3 (continued)

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Additional information

Author contributions

S.P. conceived, conceptualized and supervised the study. S.P. and B.B. designed the experiments. B.B. constructed the strains and plasmids, performed the experiments, data curation and quantitative analysis. A.D. performed image acquisition of AMR organization. B.V.T. generated the plasmids and performed yeast two hybrid screen. J.K. cloned, purified and performed the in vitro binding assay. F.C. conducted spot assays for septin kinase deletion experiment and quantification. V.R. helped with image analysis, manuscript editing and formatting. S.P., A.D. and B.B. wrote and reviewed the manuscript. All the authors were involved in editing the manuscript.

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Additional files

Supplemental Movie S1. Representative movies for the recruitment of Elm1, Gin4, Hsl1 and Kcc4 during bud emergence and cytokinesis. [🔗](#)

Supplemental Movie S2. Representative movies for the localisation and dynamics of Cdc3-mCherry in wildtype, $\Delta elm1$, $\Delta gin4$, $\Delta kcc4$ and $\Delta hsl1$ backgrounds. [🔗](#)

Supplemental Movie S3. Representative movies for the Inn1-mNG constriction in wildtype, $\Delta elm1$, $\Delta gin4$, $\Delta kcc4$ and $\Delta hsl1$ backgrounds. [🔗](#)

Supplemental Movie S4. Representative movies for the localisation and constriction of Hof1-mNG in wildtype, $\Delta elm1$ and $\Delta gin4$ during the cell cycle. [🔗](#)

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Reviewer #1 (Public review):

Summary:

The authors wanted to better understand how the various septin-associated kinases contribute to septin organization and function in budding yeast. This question has been recently addressed by similar kinds of studies but there are still some open questions, particularly as regards to what extent the kinases may interact with and/or modify components of the contractile ring that drives cytokinesis.

Strengths:

This study uses sensitive imaging with good temporal and spatial resolution to monitor the localization of various proteins in living cells. Particularly informative is the use of a GFP/GFP-binding-protein "tethering" approach to ask if the requirement for one protein can be bypassed by physically tethering another protein to a third protein. Results from a yeast two-hybrid assay for measuring protein-protein interactions *in vivo* are buttressed by direct *in vitro* binding assays using purified proteins, which is important given the likelihood of "bridging" interactions between yeast proteins in the two-hybrid approach. The authors' conclusions are quite well supported by the data.

Weaknesses:

A control for non-specific binding is missing from the *in vitro* binding assay. The figures suffer sometimes from the very small text in the labels, which obscures understanding. Ultimately, while the study provides some interesting and novel insights, we still don't understand which phosphorylation events on which proteins are important for the events occurring at the molecular level, so the advance in knowledge is somewhat incremental.

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Reviewer #2 (Public review):

Summary:

In this paper, Bhojappa et al. provide insights into the function of septin-related kinases Elm1, Gin4, Hsl1, and Kcc4 in septin organization and actomyosin ring (AMR) structure and constriction. Their findings are both corroborative of and complementary to previous related studies.

First, the authors provide a comparative analysis of the dynamic localization of these kinases at the bud neck, as well as a comparative analysis of defects in septin localization, splitting dynamics, AMR constriction rates, and cell morphology in kinase-deficient cells. They find that septin localization and splitting kinetics, as well as AMR constriction rates, are significantly perturbed in *elm1Δ* and *gin4Δ* mutants but remain largely unaffected in *hsl1Δ* and *kcc4Δ*. A similar trend is observed in terms of cell morphology and viability.

Next, the authors focus on *elm1Δ* and *gin4Δ* cells, demonstrating that the residence time of the F-BAR protein Hof1 is significantly increased and defective in these mutants. Using yeast two-hybrid (Y2H) and *in vitro* binding assays, they show that the KA1 domain of Gin4 interacts with the F-BAR domain of Hof1, which may explain the cytokinesis-related functions of Elm1 and Gin4. Supporting this, they find that Gin4's role in septin localization, AMR constriction kinetics, and Hof1 bud neck localization is kinase-independent.

The authors then conduct a series of artificial tethering experiments given their bud neck localization is mostly interdependent. They first demonstrate that artificially tethering Gin4 to the bud neck rescues the morphology defects of *elm1Δ* cells, with the strongest rescue observed when Gin4 was forced to interact with Hsl1—an effect that was also kinase-independent. Additionally, artificial tethering of Hsl1 to the bud neck restores the

morphology of *elm1Δ* cells in a KA1 domain-dependent manner, suggesting that Hsl1 functions downstream of Elm1 to maintain normal cell morphology. Consistently, artificial tethering of Elm1 to the bud neck in *gin4Δ* cells rescues morphology defects, as well as defects in Myo1 localization and AMR constriction, but only in the presence of full-length Hsl1. The rescue fails in the absence of Hsl1 or when using a version of Hsl1 lacking the KA1 domain, which supports the role of Hsl1 downstream to Elm1 in cytokinesis.

Strengths

Altogether, this study offers valuable insights into the mode of cytokinesis regulation mediated by the septin-related kinases, mainly Elm1, Gin4, and Hsl1, and would be an important contribution to the field of septins and cytokinesis after addressing current weaknesses.

Weaknesses

- (1) When assessing rescue of the *elm1Δ* phenotype, it needs to become clearer whether only morphology or also cytokinesis and septin organization are rescued.
- (2) The quantification of the microscopy data does not always match up with the example images, and it's not always clear how the authors quantitatively analyzed their data.
- (3) The forced tethering data are key to the paper, but the lack of a summarizing table makes it difficult to grasp the full picture.
- (4) Novel results and those confirming earlier results could be better distinguished.

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Reviewer #3 (Public review):

Summary:

The study by Bhojappa et al. brings new and interesting elements about the stability of the septin ring and the crosstalk between septin and actomyosin ring assemblies. The study focuses on the four kinases associated with the septin ring, Elm1p, Gin4p, Hsl1p, and Kcc4p. Elm1 and Gin4 show strong knock-out phenotypes, whereas Hsl1p and Kcc4p show weak knock-out phenotypes. The Elm1p/Kcc4p and Gin4p/Hsl1p pairs show similar timing at the bud neck. While these kinases share redundant functions, Gin4 appears to have a unique interaction with the BAR domain protein Hof1, revealing a novel direct interaction between the septin and actomyosin rings. Interestingly, the kinase activity of Gin4 is not required for its role in septin organisation and AMR constriction. The last part of the manuscript shows an original protein tethering protocol used to show that Hsl1 and its membrane binding ability are required for phenotype rescue of *gin4null* cells.

Strengths:

The combination of genetics, cell imaging, and biochemical characterization of protein-protein interactions is attractive.

Weaknesses:

- (1) Imaging and data analysis is the main weakness of this manuscript. The authors must avoid manual counting and selection when easy analysis software can be used to limit bias. Instead of presenting unclear statistics of "percentage phenotypes", they need to define clear metrics to offer meaningful phenotype analysis.

(2) This manuscript examines a very complex mechanism with four kinases of overlapping function using new data and existing literature. A clearer picture/model at the end of the manuscript that synthesizes the current knowledge would be beneficial.

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